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Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis

--Manuscript Draft--

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Full Title:	Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis
Short Title:	Predictive health scores in neonatal mouse model of sepsis
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Keywords:	neonatal sepsis; neonate; newborn; mouse model; neonatal mouse; Inflammation; infectious disease; classification; sepsis; infection; cecal slurry; machine learning; Prediction; polymicrobial sepsis; animal model, neonatal mouse model,
Abstract:	Infectious disease and sepsis represent a serious problem for all, but especially in early life. Much of the increase in morbidity and mortality due to infection in early life is presumed to relate to fundamental differences between neonatal and adult immunity. Mechanistic insight into the way newborns' immune systems handle infectious threats is lacking; as a result, there has only been limited success in providing effective immunomodulatory interventions to reduce infectious mortality. Given the complexity of the host-pathogen interactions, neonatal mouse models can offer potential avenues providing valuable data. However, the small size of neonatal mice hampers the ability to collect biological samples without sacrificing the animals. Further, the lack of a standardized metric to quantify newborn mouse health increases reliance on correlative biomarkers without a known relationship to 'clinical' outcome. To address this bottleneck, we developed a system that allows assessment of neonatal mouse health in a readily standardized and quantifiable manner. The resulting health scores require no special equipment or sample collection and can be assigned in less than 20 seconds. Importantly, the health scores are highly predictive of survival. A classifier built on our health score revealed a positive relationship between reduced bacterial load and survival, demonstrating how this scoring system can be used to bridge the gap between assumed relevance of biomarkers and the clinical outcome of interest. Adoption of this scoring system will not only provide a robust metric to assess health of newborn mice but will also allow for objective, prospective studies of infectious disease and possible interventions in early life.
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Response to Reviewers:	RESPONSE TO REVIEWER #1 Thank you for your detailed feedback and critique of this manuscript. We respond to each point raised below. •The manuscript lacks sufficient clarity to highlight the work done or the significance of it. The methods need significant work to describe what exactly was done. We have expanded the methods section to include a more detailed account of the development of the scoring system in order to highlight how this came about and have extensively reworked the section describing classifier construction. In brief, we

developed a scoring system and showed that recording these data can enable investigators to predict outcome of neonatal mice. This opens doors for the investigation of neonatal infectious disease, which is limited by an inability to collect biological samples from neonatal mice without sacrifice. We accordingly have expanded the discussion to include greater emphasis highlighting the significance of the work.

• Line 80: remove the duplicated word, “markers”
• Line 87: I think it is incorrect or overstating to state that you validated your health scoring system…
• Line 102-103: What was the volume of the IP injection into the pups?
• Line 103-104: You indicate that the litters were monitored for “morbidity (i.e. humane endpoint)” but do not describe what signs of sickness you are looking for

Typos in the manuscript have been corrected. The word ‘validated’ has been changed to ‘correlated’ to correctly reflect the data. Volume injected (~100 μL, some adjustment for weight) has been added to the methods section. The specific criteria for morbidity / humane endpoint are one of the focal points for the paper – we principally looked at righting reflex and mobility or lethargy. As this is described in much greater detail in the main text.

•What is meant by the statement “all efforts were taken to minimize animal suffering”?

As there is no standard humane endpoint or measure of sickness for neonatal mice, we feel this manuscript itself represents the claim that ‘all efforts were taken to minimize animal suffering’. We strived to identify the earliest possible point where mice that were destined to not be able to recover from disease could be identified to reduce animal suffering, and for biological importance to identify what makes those mice different from mice that go on to survive but still look sick at the same point.

•Line 116-117: please be more explicit in your description of the animals included in the analysis – were they all receiving cecal slurry? Did any receive treatment? All part of an LD50 study? All wildtype B6 mice? Did they all go to the humane endpoint or were some euthanized at 24 hours?
•Line 120: You indicate that animals were monitored 2-3 times daily throughout the experimental endpoint. What was the specific interval? Did the interval depend on the time after CS injection? What criteria determined when the animals were monitored? What time were the observations at? Light cycle or dark cycle or both?

We have edited the main text to address these questions and clarify these experiments. In survival studies mice were let to go to humane endpoint or recovery, whereas in experiments with sample collection mice were sacrificed at 24 hours. Specific intervals have been added to the methods as has the time of injection and the clarification that this occurred in both light and dark cycles.

•Please include a description of the husbandry of these animals. What sort of housing was used? What cage density? What enrichment was offered? What diet and water source were the dams offered? What was the light cycle? What was the temperature and humidity?
•Please describe how pups were identified and scored individually? How was this information tracked? Did you use scoring sheets for each animal?
•Please describe weight collection method since this data is later used as a metric for endpoint criteria evaluation

Details and clarifications addressing these questions have been added to the methods section of the text.

•Please fully describe the statistics that were used to evaluate the data? What test? Was data tested for normality? What program was used to evaluate the data?

An additional section titled ‘statistics’ has been added to the methods section in order to address these points. Some of the statistical tests used throughout the paper have been modified to better suit the data – the results have been altered accordingly.

•Figure 1: I don't understand what is plotted in A. What does the total bar at each time point represent? Why does it increase and decrease - I would expect that it would steadily decrease over time.

The total bar at each point represents the total number of mice monitored within that given bin. It does generally decrease over time, fluctuations are due to differences in monitoring time points (i.e. not many mice are monitored at 27 hours because they were just monitored at 24 hours)

•Figure 1C: were these animals all challenged with the same dose of bacteria?

All animals were challenged at a cecal slurry concentration corresponding to an LD50 – this varied slightly between different preparations of cecal slurry (from 0.8 mg-1.1 mg). The caption has been updated to better reflect this information.

•Table 1: How did you determine a 4 second cutoff time for the righting reflex? This seems like a very long time to wait for the animal to do this?

The 4 second cutoff was determined by observing the natural righting reflex of non-challenged pups, and picked as a time when over 95% of mice were able to right on at least one side after being placed on their back. Any shorter and healthy mice would display fail to right reflexes, while longer could lead to exhaustion of the mice which could potentially affect the response to disease.

•Table 1: While it is easy to understand a present or absent righting reflex, I do not follow the logic in the organization of the Mobile/Lethargic/Nonmobile associated with the scoring? Can you please explain this in the methods section – how you devised the complete scoring system.

It is difficult to give a detailed account of the development of the scoring system, as this arose from a large number of qualitative observations over the course of multiple years of running neonatal mouse experiments. The need for consistency and the ability to predict outcome drove us to turn these qualitative observations into a quantitative measurement, hence this scoring system has emerged. We have attempted to provide some background and context to address this in the methods section, and have expanded the previous discussion to include greater information about this mobility metric.

•Figure 2A: What does the shaded area represent?

The shaded area represents standard error – this has been added to the figure caption.

•Figure 2B: What is plotted in this graph and what does the shaded area represent? Are these averages or modes at each time point? A fitted model? You stated that the scores were discrete categorical data but you have a smoothed continuous curve – please explain. Can you show the actual time points for which you collected this data?

The lines represent loess fits applied to the scores treated numerically and the shaded area represents standard error. We did it was inappropriate to treat the scores as numerical variables for the purposes of statistical analysis as it is impossible to claim that the difference between each category is precisely the same arbitrary health unit. However, we felt that treating the scores as numerical entities for the sole purpose of visualizing the change in score over time was acceptable as there is no statistical test or weight being assigned to the results. We attempted to emphasize this by specifying it was a 'visual representation' – the basis of our claim that 'score changes over time' can be seen in figure S1 where the scores were not treated numerically.

•Lines 187-193: It may be useful to use a ROC curve and Youden's index to evaluate the combination of parameters you use as a cutoff – this will optimize the sensitivity and specificity of the combination of score and time point for predicting death and thus an ideal endpoint for this model.

We appreciate this suggestion for evaluation of the timepoint for predicting death, and we agree that there could be some improvement selecting the best time to predict death. However, we are somewhat limited in this manuscript by the data we have available – as most of this data was accumulated over time naturally as a result of other experiments in our lab, we were not increasing our monitoring time points to seek out the data to optimize this. So, we ultimately were presented with the choice of 12, 18, or 24 hours and here we demonstrated that, of these choices, 24 hours was the best candidate. This has been added to the discussion as a limitation of the study.

•Figure S1: Is this for all mice at 18 and 24 hours?

This is for all mice with known outcome (i.e. were not sacrificed) at 18 and 24 hours. Clarified in figure caption.

•Line 200-201: How many mice were used? Where these mice also scored using the criteria in Table 1?

125 mice were used, this information was added to the text. These mice were scored using the same criteria.

•Line 204: please clarify if the bacterial load is only assessed at 24 HPC.

Clarified in text

•Line 209-210: What data supports this statement? What health score was associated with survival? How was it correlated to bacterial load?

This text is supported by all of the data presented in previous section – Figure 2A shows a clear relationship between score at 24 HPC and percent survival and Figure 3 shows a direct correlation between increasing score and decreasing bacterial load (measured at 24 HPC) at both 18 and 24 HPC. We feel this data is clearly presented in the section.

•Line 222-223: You indicate that you use righting reflex and mobility to make a single score but then use them separately here and weight with the model. How was this information handled? Why did you separate them rather than keep them together?

Our logic in separating the score into individual components was that we wanted as close to a truly unsupervised classification process as possible – by using our created scores as inputs we worried we would limit the discriminatory capacity of the classifier. The scores were deconstructed into their smallest possible components. Mice were placed on both sides and righting reflex and mobility was collected twice so there were a total of four measurements comprising each score. Our goal was to discriminate survivors and non-survivors as effectively as possible so it logically follows to include as much information as is available. The utility of the scores external to the classifier lies more in the simplicity of collection and interpretation and thus does not make sense to separate into components.

Discussion
•Can you please discuss why you came up with score criteria but then analyzed righting reflex and body weight separately.

A brief summary of the previous response has been added to the discussion section.

•What would you specifically recommend to use as a humane endpoint criteria with this data?

As we mention in the penultimate paragraph of the discussion, we suggest that mice which fail to right and are lethargic should qualify as humane endpoint in future studies.

RESPONSE TO REVIEWER #2

We appreciate your response and critique, especially of our classification section. We have heavily modified the section to address your concerns – each point is addressed below.

	The limitations in our knowledge about the way in which newborns' immune systems handle infectious threats such as those that might lead to sepsis is correctly identified and the use of mouse models seems, in principle, like an adequate route towards increasing such knowledge. Authors define a standardized metric to quantify neonatal mouse health in its relationship to outcome (adult mouse metrics are available but not readily applicable to newborn animals. The reported interest of this metric is that it is shown to be highly predictive of survival (understood as a classification problem). The paper is well written and reasonably well structured, and the reported research falls adequately within the scope of the PLoS ONE journal. The concept of sepsis is introduced in the first section without any basic explanation or definition. Given the generalist nature of the journal, such explanation/definition would be welcome. A sentence overviewing the concept of sepsis has been added to the introduction. The description of the data analysis methods (section "Classifiers" and supplementary material) is very insufficient. Authors state that they "built a machine learning model for a binary classification problem", but the fact is that only part of their classifiers could be considered to be machine learning models. More importantly, there seems to be no clear justification or rationale behind the specific choice of algorithms. Such justification should be made explicit in the text. We have heavily modified the classifier section and re-wrote the supplementary material entirely to try to improve clarity and provide more background that may be required for this manuscript to flow naturally. The new supplementary section is primarily focused on the differences between the approaches and the logic behind our choices. Additionally, we have created a public GitHub repo wherein the code can be reviewed which should clarify any specific questions (S1 URL). All these algorithms depend on a number of parameters that must be tuned to create the classifier model. No reference is made to how the optimal values for these parameters were selected. Without such information, the experimental results have no guarantee of replicability. Authors should also clarify how they tackled the potential problem of overfitting in their models. The evaluation metrics (accuracy and AUC), even if standard, should be formally described and justified. The feature selection methods are barely described (for instance, what is KNN and SV classification feature selection?) We have added an additional table in the supplementary information which outlines the user-defined parameters which needed to be defined in order for the algorithms to progress. We principally used the GridSearch technique, which we have now defined and included in the manuscript. Further, we have added a discussion of the cost / benefits of each algorithm and provided some rationale behind the choices made. Additionally, we have uploaded our Python workbook which details the whole code and should address any concerns about repeatability. We have also now clarified accuracy and AUC in the main text. In the discussion, I miss a couple of things: 1) comment on the relevance of feature selection as part of the classification analysis and as a way to make the interpretation of the results more accessible to medical experts; and 2) some comments on how this preliminary research on mouse models could eventually lead towards an approach that might be useful in human neonatal research. The relevance of feature selection has been added to both the results and the discussion, emphasizing the significance of righting reflex and its quality as an easy to use, predictive metric. We have also added some additional lines in the discussion highlighting the relevance to human neonatal infectious disease. Figures: All figures seem to be of rather low resolution (don't know about the originals, but they look so in the pdf file)

Resolution has been fixed on figures and some have been removed.

Confusion matrices in Figs.4 and S3 do not make much sense as figures. They would be better reported as tables.

The confusion matrices have been replaced with tables.

Several figures in the supplementary material come with no descriptive text.
I do not quite understand what the horizontal axis represents in Fig.S2

We have updated some figures which erroneously did not include axes labels. Per the PLoS ONE style requirements, supplementary figures are not required to include descriptive text – those which do not have this text are referenced in the supplementary materials and are meant to be understood in the context of that document.

Data are adequately made publicly available from the Open Science Framework database.

RESPONSE TO REVIEWER #3

Thank you for your detailed review of our manuscript – your feedback has been very helpful in guiding us to clarify some confusing points. We address each of your concerns below and have added additional supplementary information.

Neonatal mouse could be utilized as a model and provide valuable data for the research of sepsis in newborn babies. Due to the significant hardness in collecting biological samples in neonatal mouse without sacrificing the animals, new methods based on easy-obtained quantitative and qualitative metrics beyond the regular adult predictors are essential and should be established.

In this manuscript, the authors successfully presented some critical new metrics and constructed a scoring system to indicate the health and bacterial load of neonatal mouse. In addition they constructed a classifier to predict the mortality and validated the robustness.

Generally this research is quite innovative and produced good and meaningful results.

But there are some questions/problems to be answer/clarify and some flaws to be corrected:

1) Authors should demonstrate the bacterial load but not the IP injection surgical damages and related stress is the main reason of changes in righting reflex.

We have inserted an additional supplementary figure showing that sham challenges (injection of either dPBS or dextrose 5% water) did not cause any mortality (A) and had significantly higher scores than mice which received a cecal slurry challenge (B).

2) Prediction accuracy alone is NOT a good indicator of prediction performance as it's hardly to be extended to the usually highly unbalanced real world data.. Authors could replace it with (macro or micro) F1 score, because it much more robust.

We entirely agree with the reviewer's sentiment that accuracy is a poor metric of performance in real world (unbalanced) data, but we believe it is appropriate in this context where we do in fact have a highly balanced set (112 survivors and 110 non-survivors). Still we did look at F1 scores and found virtually no difference compared to accuracy – these can be seen in the now posted code associated with classifier construction. Again, we entirely agree with the reviewer, but our inclination is to keep accuracy as the presented metric in the paper for simplicity and readability, as the intended audience is experimental scientists rather than experts in machine learning. This classifier is not intended to extend to real-world data - its utility lies in the ability to discriminate survival in a controlled laboratory setting where it is unlikely to encounter highly imbalanced groups.

3) Line 173:
Authors should show all data about the timing of mice death, no matter what's the prediction results. If the death was long after the scoring, it’s could caused by factors other than the infections and it hard to validate effectiveness of scoring/prediction. The autopsy data could validate the method if they are available.

We agree with the reviewer that all deaths are important regardless of the timing, and for all statistical analyses (and classification) we indeed included all deaths. Specifically for Figure 2B, the loess fit shown no longer makes sense as the data gets extremely sparse after 48 hours. The monitoring time points become less frequent for the survivors (as they are already in recovery) and so the majority of the recorded scores are mice which continue on to death, as these mice are still being monitored frequently. All mice have reached recovery or death by 96 HPC so it seems unlikely to us that these few deaths after 48 HPC are unrelated to the infection. Unfortunately the autopsy data are no longer available for this dataset.

4) Line 230:
Authors should clearly demonstrate what's the initial features and the rational of the initial features composition.

An overview of the feature inclusion has been added to the methods section – the rationale was to include all information available that was present in this data set and then employ a rigorous, data-driven method of selecting the most relevant features. This data set is comprised of mice from many experiments over a multi-year span which happened to be treated the same (i.e. received a cecal slurry challenge at the correct dose and were not exposed to any sort of treatment or intervention). This paper and classifier represents us working with the data which we had been collecting already rather than an experiment-driven approach to defining neonatal mouse health. We realized these data we had been collecting were very strong in of themselves and so set out to create a scoring system which can be adopted by other labs working with neonatal mice. So in summary, the rationale behind the initial feature selection is that these were things we had qualitatively observed to be relevant to survival and therefore were worth collecting. We have attempted to make this more clear in the manuscript.

5) Line 233:
Why Pearson correlation coefficients were used to select features and what's the threshold ? why choose the threshold?

Information about the threshold (0.2) has been added to the methods. Similarly to hyperparameters optimization, when we do correlation and decide to select feature we need to have a method that “tells us” or suggest us the threshold to use to have, along with the hyper parameters optimized, the best combination for the best model accuracy, thus the choice of the best cutoff to use to pick our features is optimized and not based on an arbitrary cutoff choice.

6) Line 237:
Why only 74 cases to be used to classify as survivor/non-survivor? In fact the number changed everywhere (in line 117, in "CFU_with_predicted_outcome.csv", in "Mortality_data_training_and_test.csv", and in Fig 1B -- total mice is 138+128 = 266)

We have attempted to clarify the confusion regarding the mouse numbers by overviewing the numbers more explicitly in the methods section. Out of 266 mice, 21 were saved for an external validation and 23 had missing values that made them ineligible to predict outcome. These mice were still able to be used in the direct comparisons of score / righting reflex and mortality. 222 mice were used to train and test the classifier, which meant these were further subset into to categories: 66% were for training (148) and 33% (74) were for testing. This is where the 74 comes from. We have also added another supplementary document outlining the datasets and the number of mice within each in order to help alleviation any potential confusion.

7) Line 247:
What's biometric data? Authors should clarify.

Biometric data meaning weight and sex but principally weight – we agree this is a confusing term and have adjusted the figure title accordingly.

8) Line 257 and line 275:
The surgical damage and operation stress could cause morbidity, authors should exclude the effects of surgical damage and operation stress or at least clearly discuss it in detail.

We have addressed this with an additional supplementary figure (discussed in answer to question 1).

9) Line 392:
Where is the details/setting/parameters about the Method: GridSearch and combination of GridSearch with others methods?

We have now uploaded the code used to construct this classifier and have directed readers there if they would like to see the specific parameterization used. We agree that these parameters are critical for repeatability, but we feel it is beyond the scope of this paper to discuss the settings of GridSearch in detail given the intended audience of experimental scientists.

10) In the data of "CFU_with_predicted_outcome.csv"
10.1) Why there are only 164 mice info? It's different from 404 mice authors claimed in line 117
10.2) Why are there many NA/missing value in the column "Probability.Live"?
10.3) What's the method to produce the data in the column "Probability.Live"?
10.4) Irrelevant data (columns) should not be included

Thank you for your feedback on these data. This dataset represents a subset of mice which were sacrificed at 24 hours and the bacterial load collected. The missing values in probability live were mice that were erroneously included in this data set and were not represented in the paper – these mice have been removed and irrelevant columns have been cleaned. To alleviate potential confusion, we have added an additional supplementary document clarifying the contents of each file which has been uploaded. We have also added a glossary tab to each data frame attached to the manuscript, per your later suggestion.

11) In the Fig. S4:
11.1) Why "v18.hour.post.challenge" "v24.hour.post.challenge" used to check correlation? It looks pointless
11.2) What's the meaning of "chal.status_non_challenged" ?
In fact, every headline column/item should be explained or have glossary of the definition,
11.3) The rationale these items are included should be explained

These columns show the exact time post-challenge that mice were monitored. All features were checked for correlation to ensure an unbiased assessment – we agree it was highly unlikely that the very slight differences in time post challenge would make a difference, but it is important to not make assumptions when it can be easily confirmed by the machine. This indicated whether mouse received a cecal slurry or sham challenge – two mice in the set were unchallenged and not excluded. A glossary has been added to an additional tab in each data set to clarify the meaning of column names and values within.

12) In the data "Mortality_data_training_and_test.csv"
12.1) Why there are 299 mouse info? It's different from 404 mice author claimed in line 117
12.2) how to treat or imputed the missing data in the above tables?
12.3) Irrelevant data (columns) should not be included.

	This data set represents all of the mice for which mortality data was collected – again, some mice were incidentally included which had been filtered out of the final paper due to missing data (hence the prevalence of missing data here). These have been removed and two additional subsets have been added for clarity. Irrelevant columns were again removed. 13) Table S3: Where is the label of Y-axis in Fig S3 B? 14) Table S5: Where is the label of Y-axis? The labels have been added to the axes, thank you. 15) Where is the information about the programming language/ softwares/ libraries ? 16) Where is the programming code? In response to both questions 15 and 16 - the programming code has been uploaded to a public github repository and a link has been provided in the supplementary information. For ease of access it is also linked here: https://github.com/radaniba/Sepsis_Project 1) Line 80, two repeated "markers" 2) Line 139, 140, 145 -- Fig 1A, Fig 1b, Fig 1c; Either up case or lower case, but keep consistent. Thank you for identifying these errors, they have been corrected in the manuscript.
Additional Information:	
Question	**Response**
Financial Disclosure Enter a financial disclosure statement that describes the sources of funding for the work included in this submission. Review the submission guidelines for detailed requirements. View published research articles from *PLOS ONE* for specific examples. This statement is required for submission and **will appear in the published article** if the submission is accepted. Please make sure it is accurate.	This research was supported by funding from the Canadian Institute for Health Research (CIHR). T.R.K. is supported by a Michael Smith Foundation for Health Research Career Investigator Award. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

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Competing Interests Use the instructions below to enter a competing interest statement for this submission. On behalf of all authors, disclose any competing interests that could be perceived to bias this work—acknowledging all financial support and any other relevant financial or non-financial competing interests. This statement **will appear in the published article** if the submission is accepted. Please make sure it is accurate. View published research articles from *PLOS ONE* for specific examples.	The authors have declared that no competing interests exist.

NO authors have competing interests Enter: *The authors have declared that no competing interests exist.* **Authors with competing interests** Enter competing interest details beginning with this statement: *I have read the journal's policy and the authors of this manuscript have the following competing interests: [insert competing interests here]* * typeset	
Ethics Statement Enter an ethics statement for this submission. This statement is required if the study involved: • Human participants • Human specimens or tissue • Vertebrate animals or cephalopods • Vertebrate embryos or tissues • Field research Write "N/A" if the submission does not require an ethics statement. General guidance is provided below. Consult the submission guidelines for detailed instructions. **Make sure that all information entered here is included in the Methods section of the manuscript.**	Male and female C57BL/6 mice aged 6 weeks (Jackson Labs, catalogue #664) were purchased from The Jackson Laboratory and used for breeding purposes; all mice used in this study were first or second-generation progeny of Jackson mice bred in-house. This study was carried out in strict accordance with animal ethics guidelines outlined by University of British Columbia and was approved by the Animal Care Committee (protocol numbers A17-0110 and A14-0261). Animals were housed in OptiMice cages, with food and water ad libitum. Breeding mice were fed a high fat diet (Tekland Cat 2919) while others were fed a low-fat diet (Tekland Cat 2918). The pellet base was cellulose ¼" Performance Bedding (Lab Animal Supplies INC, Cat L0108), and environmental enrichment included a red hut, nesting material of Envirodri (Sic.) (Jameson, B501) and Nestlets (Ancare, Cat CABFM00088). Mice underwent a controlled 12 hour light cycle, with temperature maintained at 23 °C. Mice were anesthetized with isoflurane prior to euthanasia through CO_2 exposure, confirmed by decapitation. Humane endpoint was defined as the lack of an attempt to right themselves when placed on their back on both sides

Format for specific study types **Human Subject Research (involving human participants and/or tissue)** • Give the name of the institutional review board or ethics committee that approved the study • Include the approval number and/or a statement indicating approval of this research • Indicate the form of consent obtained (written/oral) or the reason that consent was not obtained (e.g. the data were analyzed anonymously) **Animal Research (involving vertebrate animals, embryos or tissues)** • Provide the name of the Institutional Animal Care and Use Committee (IACUC) or other relevant ethics board that reviewed the study protocol, and indicate whether they approved this research or granted a formal waiver of ethical approval • Include an approval number if one was obtained • If the study involved *non-human primates*, add *additional details* about animal welfare and steps taken to ameliorate suffering • If anesthesia, euthanasia, or any kind of animal sacrifice is part of the study, include briefly which substances and/or methods were applied **Field Research** Include the following details if this study involves the collection of plant, animal, or other materials from a natural setting: • Field permit number • Name of the institution or relevant body that granted permission	
Data Availability Authors are required to make all data underlying the findings described fully available, without restriction, and from the time of publication. PLOS allows rare exceptions to address legal and ethical concerns. See the PLOS Data Policy and FAQ for detailed information.	Yes - all data are fully available without restriction

A Data Availability Statement describing where the data can be found is required at submission. Your answers to this question constitute the Data Availability Statement and **will be published in the article**, if accepted. **Important:** Stating 'data available on request from the author' is not sufficient. If your data are only available upon request, select 'No' for the first question and explain your exceptional situation in the text box. Do the authors confirm that all data underlying the findings described in their manuscript are fully available without restriction?	
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and contact information or URL). • This text is appropriate if the data are owned by a third party and authors do not have permission to share the data. * typeset	
Additional data availability information:	

Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis

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Abstract

Infectious disease and sepsis represent a serious problem for all, but especially in early life. Much of the increase in morbidity and mortality due to infection in early life is presumed to relate to fundamental differences between neonatal and adult immunity. Mechanistic insight into the way newborns' immune systems handle infectious threats is lacking; as a result, there has only been limited success in providing effective immunomodulatory interventions to reduce infectious mortality. Given the complexity of the host-pathogen interactions, neonatal mouse models can offer potential avenues providing valuable data. However, the small size of neonatal mice hampers the ability to collect biological samples without sacrificing the animals. Further, the lack of a standardized metric to quantify newborn mouse health increases reliance on correlative biomarkers without a known relationship to 'clinical' outcome. To address this bottleneck, we developed a system that allows assessment of neonatal mouse health in a readily standardized and quantifiable manner. The resulting health scores require no special equipment or sample collection and can be assigned in less than 20 seconds. Importantly, the health scores are highly predictive of survival. A classifier built on our health score revealed a positive relationship between reduced bacterial load and survival, demonstrating how this scoring system can be used to bridge the gap between assumed relevance of biomarkers and the clinical outcome of interest. Adoption of this scoring system will not only provide a robust metric to assess health of newborn mice but will also allow for objective, prospective studies of infectious disease and possible interventions in early life.

Introduction

An estimated 3.0 million cases of newborn infection every year result in nearly 350,000 deaths, making infection one of the most lethal threats in early life [1]. Many deaths due to infection can be attributed to sepsis, a difficult to define disease characterized by failures of regulatory immune system components resulting in hyper- or hypo-inflammation [2,3]. It is widely known that the neonatal immune system is fundamentally distinct from that of adults, indicating that data generated from adult animal models should not be assumed to apply to newborns [4–7]. While neonatal mouse models offer several practical benefits (availability, existing infrastructure, well-understood biology, etc.), working with neonatal mice presents some unique challenges which have not yet been effectively addressed. In particular, researchers are limited by the size of the mice; day of life (DOL) 7 mice (considered to be immunologically closest to human term newborns at birth) [8] are still far too small to collect meaningful volumes of blood without sacrificing experimental animals. In adult mice, about 7 to 8 µL of blood can ethically be collected per gram of mouse bodyweight without sacrifice or fluid replacement – this translates to a maximum single collectible blood volume from neonatal mice of about 20 µL [9]. Serial sampling (every 24 hours) would only allow for collection of 5 µL of blood or less in each draw [10]. This limitation (in addition to the practical difficulties of drawing blood from such small animals) only allows terminal methods of bleeds to collect samples of adequate volume, which fundamentally separates biological findings from true 'clinical' outcomes (i.e. survival). This inability to serially sample neonatal mice without sacrifice also forces researchers to rely heavily on biomarkers as a stand-in for outcome when exploring a potential treatment of interest, assuming causality between e.g. pathogen load or inflammatory cytokine production and mortality.

Health scores for sepsis are widely used in human clinical settings as they are critical for capturing disease progression and can inform potential interventions [11–15]. The ability to assess disease severity in a quantitative way is also extremely useful in animal models as it can help investigators avoid an overreliance on assumptions about the relevance of a given biomarker. For example, one group developed a series of symptom scores for adult mice in a similar model of sepsis (fecal suspension injected IP) which relied on, among other things: activity, response to stimulus, eyes (open or shut), level of consciousness, and appearance [16]. This study illustrated that when grouping adult mice by these health scores, TNFα and IL-1β levels did not correlate with outcome despite elevated levels in comparison to unchallenged controls – these cytokines went up following challenge but did not appear different in survivors and non-survivors. This demonstrates the capacity for health scores to narrow the focus of investigators onto biomarkers which are functionally relevant to a clinical outcome of interest – a capacity even more critical when working in a neonatal population from which even blood samples are extremely difficult to collect without harming them. While the quantitative observations used in this model may be strong markers for adult animals, none of these characteristics are readily applicable to a neonatal population (i.e. no fur to be ruffled, eyes are always closed, minimal basal activity levels, etc.).

To address this need, we developed a data-driven and simple scoring system directly related to outcome (humane endpoint) in neonatal mice. The addition of our health scores to neonatal mouse experiments provides: (a) an improved ability to track disease progression which can inform time of sacrifice for sample collection and (b) a quantitative method to prospectively differentiate survivors from non-survivors. We here correlate this health scoring system with mortality in a polymicrobial model of neonatal sepsis. This model confirms that clinical outcome was highly correlated with bacterial load. It took less than 20 seconds to determine these composite health scores per animal. The prospective validity of our health score in assigning outcome provides the much-needed metric to begin assessing causality of biomarkers, which in turn will allow assignment of mechanistic cause-effect relationships. This approach will not only decrease undue suffering of the animals, but also increases the likelihood of identifying effective immunomodulatory interventions for infectious disease and sepsis.

Materials and Methods

Mice

Male and female C57BL/6 mice aged 6 weeks (Jackson Labs, catalogue #664) were purchased from The Jackson Laboratory and used for breeding purposes; all mice used in this study were first or second-generation progeny of Jackson mice bred in-house. This study was carried out in strict accordance with animal ethics guidelines outlined by University of British Columbia and was approved by the Animal Care Committee (protocol numbers A17-0110 and A14-0261). Animals were housed in OptiMice cages, with food and water *ad libitum*. Breeding mice were fed a high fat diet (Tekland Cat 2919) while others were fed a low-fat diet (Tekland Cat 2918). The pellet base was cellulose ¼" Performance Bedding (Lab Animal Supplies INC, Cat L0108), and environmental enrichment included a red hut, nesting material of Envirodri (Sic.) (Jameson, B501) and Nestlets (Ancare, Cat CABFM00088). Mice underwent a controlled 12 hour light cycle, with temperature maintained at 23 °C. Mice were anesthetized with isoflurane prior to euthanasia through CO_2 exposure, confirmed by decapitation. Humane endpoint was defined as the lack of an attempt to right themselves when placed on their back on both sides, explained in greater detail below.

Cecal slurry model of neonatal sepsis

Polymicrobial sepsis was induced using a modified version of the model first described by Wynn *et al.* in 2007 and recently outlined on video in the Journal of Visualized Experiments [17,18]. Briefly, cecal material was extruded from adult male mouse ceca (aged 6-10 weeks), diluted in dextrose 5% water (D5W) to 160 mg slurry / mL D5W, filtered through a 70 µm sterile strainer, pooled, aliquoted, and frozen at -80 °C. Aliquots were thawed at room temperature, diluted further in D5W to the desired weight-adjusted dose, and kept on ice prior to intraperitoneal (IP) injection. The challenge dose was titrated to achieve a lethal dose equal to 50% (LD50) per litter. Neonatal mice were injected IP with approximately 100 µL cecal slurry (0.8 – 1.1 mg/g mouse) on day of life (DOL) 7 or 8 to induce polymicrobial sepsis, with slight variations in volume due to weight adjustment. Litters were monitored for morbidity (i.e. lethargy, inability to right) up to 96 hours post challenge (HPC). In these survival experiments, mice were monitored either until full recovery was observed (gaining weight, alert / active), or until humane endpoint was reached at which point the mice would be euthanized. A separate cohort of pups were sacrificed at approximately 24 HPC in order for the collection of blood and tissue samples. Bacterial loads in liver, lungs, spleen, and blood were subsequently calculated by serially diluting the organ homogenates in PBS, drop-plating on 5% sheep's blood agar, and the counting colony forming units (CFU) after 24 hours of incubation at 37 °C.

Monitoring and health scores

Monitoring commenced 12 HPC (i.e. prior to the onset of expected mortality) and continued 2-3 times daily (07:00 – 09:00, 12:00-14:00, 16:00-18:00, spanning both light and dark cycles) through to experimental endpoint. All mice were challenged in the evening (17:00 – 18:00) so monitoring schedules were not adjusted based on time of challenge. During mouse scoring the neonatal mice were separated into a secondary cage to avoid stressing the dam. Nesting material was rubbed in hands to transfer that litter's smell to the gloves, and then each mouse was individually scruffed and placed on their back as visualized in Brook *et al.* 2019 [18]. Health scores arose naturally out of a need to define a humane endpoint for extremely ill neonatal pups. Placing a pup on its back and recording whether it was able to right itself was a simple and effective discriminator of outcome, but alone it failed to capture the obvious difference between mice which were vigorously attempting to right and those which barely moved. Thus, we began recording qualitative observations of vigor to pair with righting reflex data, which eventually became quantitative descriptors of mobility (i.e. angle of hip movement when attempting to right). Righting reflex measurements were further standardized by adding a four-second cutoff, determined by the observation that more time was unnecessary, and less time risked inappropriately assigning 'fail to right' to a healthy pup. For simplicity of recording and interpretation, these observations were transformed into ranked scores from 0 - 5, as presented in Table 1. Scores were recorded at each monitoring time point in duplicate. In addition to scoring, mice were weighed at each time point using a top-loading balance. All mice used were either 7 or 8 days old and weighed an average of 3.7 g (2.25 g - 6.06 g). Pups were identified by markings with an ethanol proof pen on the top or bottom of the tail. Monitoring sheets for each litter were used to record scores and weights. A total of 424 mice were used in this study: 266 with an endpoint of mortality or recovery, 125 sacrificed at 24 hours for a readout of bacterial load, 33 used in follow-up experiments. In survival experiments, all mice were allowed to proceed to humane endpoint or recovery (gaining weight four days post-challenge with scores of 4-5), whereas in bacterial load experiments all mice were sacrificed at 24 HPC regardless of perceived illness.

Classifier

The ability to confidently predict outcome (survival or non-survival) at time of sacrifice would greatly help limit the number of assumptions required by investigators. We interpreted survival vs. non-survival as a binary classification problem and constructed multiple machine learning models aimed at classifying mice into one of these two groups at 24 HPC. In order to identify the best model, we tested our dataset in 6 different algorithms split across two primary classes: baseline learning algorithms (Logistic Regression, K Nearest Neighbors and Decision Tree) and ensemble learning algorithms (Random Forest, Gradient Boosting and XGBoost). Feature selection began from all available data collected for each mouse: sex, date of birth, time of challenge, weight at challenge, weight at each monitoring time point, precise HPC each monitoring time occurred, righting reflex, mobility at each monitoring time point, and the change in any of the above from one point to another. Pearson correlation coefficients were used to avoid including highly co-correlated and irrelevant features, with a correlation threshold of 0.2 (with outcome) acting as the minimum cutoff for inclusion. This threshold and all other parameters were optimized iteratively using GridSearch, which examines all possible combinations of values for each parameter and tunes each parameter to maximize classification strength. A more detailed description of the other approaches we examined for feature selection is presented in the supplementary material (S1 Appendix).For specific parameters used during classifier construction, see the source code linked in the supplementary materials (S1 URL). The righting reflex and mobility were treated separately (rather than combined as scores) to maximize the amount of information available for the classification problem and minimize the potential for biasing what should be an unsupervised process with our own assumptions about what should be associated with mortality. A total of 222 mice were split into a training set (148 mice) and a test set (74 mice) and the different approaches were evaluated by accuracy (total number of correct classifications over total number of instances) and area under the receiver operating characteristic curve (AUC), which quantified sensitivity and specificity and represented a good metric of the quality of a classifier (S1 Table). Evaluation based on these performance metrics led us to select Gradient Boosted Machine learning as the strongest classification algorithm and was applied to an external data set of 21 mice for further validation. A detailed description of classifier construction and background behind the various approaches examined are available in the supplementary material (S1 Appendix).

Statistics

Comparisons of survival curves were evaluated using the log-rank test, with statistical significance assigned when $p < 0.05$. All tests comparing score and survival excluded the assigned score of 0, as this was the defined humane endpoint and thus could not be treated as independent. Percent mortality was analyzed using two-sided Fischer's exact tests with Bonferroni adjustment where appropriate. Scores were only treated as numerical quantities for the purpose of examining change over time – all other analyses treated scores as categorical. Any statistical test applied identically to blood, spleen, liver, and lung was adjusted using the Bonferroni method. Bacterial load data did not exhibit a normal distribution (per Shapiro-Wilk normality test) and were compared using the Wilcoxon rank-sum test. When analyzing independent variables with multiple levels, data were first tested with the non-parametric Kruskal-Wallis test and only if significant ($p < 0.05$) were post-hoc, pairwise comparisons made using the Wilcoxon rank-sum test (Benjamani-Hochberg adjusted). All statistical analyses were performed in R (version 3.5.4).

Results

Health scores and outcome

At all monitoring time points, righting was associated with survival and failure to right (FTR) was associated with non-survival (Fig 1A). As 24 HPC was as far into disease progression as one could wait to sacrifice mice without incurring heavy survivor bias (Fig 1B), this timepoint became the time of sacrifice for the alternative cohort of mice assessed for bacterial load. It was therefore critical to be able to distinguish survivors and non-survivors prior to sacrifice and no later than 24 HPC. Mice which failed to right at 24 HPC were significantly more likely to die than those which successfully righted (log-rank test, $p < 0.001$) and had a statistically significant, 100-fold higher bacterial load across all compartments (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$) (Fig 1C). Righting reflex alone was strongly correlated with survival and bacterial load, especially after 24 HPC. We also captured three different levels of mobility: non-mobile, lethargic, and mobile (Table 1). One of the strengths of a scoring system should be the ability to provide a metric predictive of future outcome. To assess the predictive value of this righting / mobility combination metric, we assigned scores from 0 (fail to right non-mobile, humane endpoint) to 5 (rights mobile, perfectly healthy) (Table 1). Unless otherwise indicated, the reported score is the lower of the two recorded for each mouse.

Figure 1. Righting reflex at 24 HPC is an excellent discriminator of survival and bacterial load. Righting reflex of neonatal mice (DOL 7-8) challenged IP with cecal slurry at an LD50. (A) Ability to right at different hours post challenge (HPC) in survivors and non-survivors. (B) Survival curve of mice split by righting reflex at 24 HPC. Mice which were able to right themselves at 24 hours were significantly more likely to survive the cecal slurry challenge than mice which failed to right (log-rank test, $p < 0.001$). (C) Bacterial load split by righting reflex at time of sacrifice, 24 HPC. Mice which were able to right had significantly lower bacterial load in blood and all measured organ tissues (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$).

Table 1. Assigning numerical health scores to quantitative observations in neonatal mice

Numerical Score	Righting Reflex*	Mobility	Description
5	Rights	Mobile	• Multiple steps in a row, responsive to tail tap • Alert, may be shaky but moving with energy • Might fall over while walking
4	Rights	Lethargic	• Takes at least one step, but stops between steps • Steps are slower, may be very shaky, may fall over

3	Rights	Nonmobile	• Stationary after righting • May take a half-step after tail tap but largely or entirely unresponsive
2	Fails to right	Mobile	• Vigorous hip movement, swinging side to side at an angle > 90° from horizontal at least once during the observation period (4 seconds)
1	Fails to right	Lethargic	• Slower hip movement, < 90° angle • Halting attempts to right over the observation period, may be some pauses but always starting again
0	Fails to right	Nonmobile	• Zero or extremely weak hip movement • Gives up attempting to right or never starts, may be uncontrollably shaky • limbs may bend at the knee and extend but hips will not rock side to side

*Righting reflex measured by placing mice gently on their back and observing their ability to right within 4 seconds. All scores were taken in duplicate with the lower of the two reported here.

Health scores are strongly associated with survival and bacterial load

Combining mobility with righting reflex allowed us to generate six categories which were assigned to numerical scores for ease of recording and visualizing. As the score of zero was not independent from mortality (score of zero was the defined humane endpoint and necessitated euthanasia), it was excluded from statistical analyses. Mice which received sham challenges with either phosphate buffered saline (PBS) or dextrose 5% water (D5W) exhibited no mortality and had significantly higher scores than those which received cecal slurry, indicating that the drop in score post-challenge was not a reflection of injury due to the injection itself but rather was in response to disease (S1 Fig). Comparing the categorical scores at 18 and 24 HPC indicated a significant relationship between score and survival (Fisher's exact test, $p < 0.001$). A simple linear regression model examining percent overall survival against assigned score at 24 HPC (mice with a score of 0 were excluded) was found to be significant ($R^2 = 0.946$, $p = 0.0035$, $n = 210$) with each increase in score equating to a 22% ± 8% improved chance of survival (Fig 2A). Visualizing the changes in score over the first 48 hours of disease (capturing 93% of deaths) showed a clear separation between survivors and non-survivors beginning around 18 HPC, with higher scores associated with recovery and survival (Fig 2B).

Figure 2. Health scores were directly associated with outcome in a polymicrobial model of sepsis in neonatal mice.

(A) Scores recorded at 24 hours post challenge (HPC) plotted against the percent of mice which survived the IP cecal slurry challenge. Scores of zero were excluded as they were the pre-determined humane endpoint. A significant linear regression was found (R^2 = 0.946, p = 0.0035, n = 210) with percent survival increasing by 22 ± 8% for each single point increase in score. Shaded area represents standard error. (B) Visual representation of change in score defined numerically over time following IP cecal slurry challenge.

The inability to distinguish survivors through health scores at 12 HPC (Fig 2B) was consistent with qualitative observations that most mice appeared similar during this time period irrespective of final outcome. At 24 HPC, the scores were well distributed with the least frequent score (3) assigned to 10.9% of mice (29/266) and the most frequent, the healthiest score of 5, assigned to 21.0% (56/266) of mice (S2 Fig). At 18 HPC, many of the mice appeared similarly ill, with only 6.9% at the healthiest score of 5 and 46.8% at the middle score of 2. Thus, health scores at 24 HPC were most useful in their ability to capture a range of sickness in neonatal mice.

To test the validity of our scoring system, a separate cohort of 125 mice generated under the same experimental conditions were sacrificed at 24 HPC and samples of whole-blood, spleen, liver, and lung tissue were plated on sheep's blood agar and the colony forming units (CFU) growth was assessed 24 hours later. Two separate, non-parametric Kruskal-Wallis tests were performed to assess the relationship between categorical scores at 18 and 24 HPC and bacterial load (at 24 HPC) across all four biological samples. A significant effect was attained both at 18 HPC (Bonferroni-adjusted p-values < 0.001) and 24 HPC (Bonferroni-adjusted p-values < 0.001) (Fig 3). The distribution of scores broadened as disease progressed towards an outcome – nearly all mice scored between 1-3 at 12 HPC, most fell within that same range at 18 HPC, but an even distribution was present at 24 HPC (S2 Fig). Our health scores were strongly associated with both survival and bacterial load at 24 HPC.

Figure 3. Health scores at 18 and 24 HPC are related to bacterial load at 24 HPC.

(A) A Kruskal-Wallis comparison of score 18 hours post challenge (HPC) with bacterial load in whole-blood and tissues showed a significant effect (Bonferroni-adjusted p-values < 0.001) indicating score at 18 HPC was related to the bacterial load measured in tissues of the same animals collected 6 hours later (24 HPC). (B) A significant effect was also attained at 24 HPC (Bonferroni-adjusted p-values < 0.001) demonstrating the relationship between health scores and bacterial load across all biological compartments.

Righting reflex and mobility are predictive of outcome prior to sacrifice

We used a machine learning approach to classify mice as survivors or non-survivors based primarily on righting reflex, mobility, and weight, specifically looking at how each metric changed from 18 to 24 HPC. Only 12% (28 / 222) of mice were at humane endpoint at 24 HPC, so while the association between score components and outcome may be influenced by the inclusion of these mice (mice which were FTR non-mobile on both sides are at humane endpoint and must be sacrificed), they did not represent a large enough proportion to call the results into question. Pearson correlation proved the most effective method of feature selection and identified attributes most associated with death: a mobility score of "non-mobile", a change from able to right at 18 HPC to failed to right at 24 HPC, and weight loss between 18 and 24 HPC (Fig S4). The ability to right was strongly correlated with survival, especially the consistent ability to right at both 18 and 24 HPC.

We used 10-fold cross validation training on a combination of six algorithms and three methods of feature selection in order to identify the most effective method of classification (S1 Table). A more detailed description of this approach is available in the supplemental section appendix 1 (S1 Appendix). For the final predictions, we used Gradient Boosting Machine as our classifier, combined with features selected by Pearson correlation coefficients. This approach had a cross validation average accuracy of 0.91 with a low standard deviation of 0.06, indicating little risk of over or underfitting (S3 Fig). With the hyperparameters selected for our Gradient Boosting model, we built a model on the training data subset with known outcome and applied it to 33% of the original data set not used during the training phase. With a total of 74 cases to classify as "survivor" or "non-survivor", the Gradient Boosting model performed well with an accuracy score of 0.85 and an AUC of 0.93 (Fig 4A), with 33 true positive cases classified as "non-survivor" and 30 cases classified as "survivor" (Table 2). To further validate our model, we generated an additional dataset with the same data collection process as well as the same data cleaning and standardization pipeline. With a total of 21 new data points, our model accurately classified 18 with an average score of 85% accuracy (S2 Table). Righting reflex and mobility at 24 hours together with change in weight were the features most strongly correlated with outcome (S3 Figure). Finally, we applied our classifier to aa different set of pups sacrificed at 24 HPC and identified a clear relationship between survival and bacterial load (Fig 4B).

Table 2. Confusion matrix showing accuracy of Gradient Boosting Machine model when applied to test set of 74 mice. Using monitoring data collected at 18 and 24 HPC (weight, righting reflex and mobility), a Gradient Boosted machine learning model was able to distinguish survivors and non-survivors with an accuracy score of 0.85; 33/39 survivors and 30/35 non-survivors were correctly classified.

True class	Predicted class: Survivor	Predicted class: Non-Survivor
Survivor	**33**	6
Non-survivor	5	**30**

Figure 4. Survival or non-survival of neonatal mice was predicted at 24 HPC and was strongly associated with bacterial load.

(A) Receiver Operating Characteristic curve illustrating the sensitivity and specificity of a Gradient Boosting Machine model trained on 148 mice and tested here on 74 mice. The model was primarily driven by weight change, righting reflex and mobility at 18 and 24 HPC. (B) Neonatal mice injected IP with cecal slurry were sacrificed 24 hours post challenge, with tissue and blood samples plated on sheep's blood agar at time of sacrifice. Mice were predicted to survive or not survive based on righting reflex, mobility, and other biometric data. Mice which were predicted to survive had a significantly lower bacterial load in all measured biological compartments than predicted non-survivors (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$). All mice with zero bacterial load in the whole-blood (n=9) were predicted to survive.

Discussion

Here we present a system of health scores for neonatal mice which were used to confidently predict mortality 24 HPC in an established model of neonatal sepsis [17]. The significance of this is twofold: 1) the classifier enables investigators to include "clinical" outcome (e.g. mortality) as a covariate of interest when all animals are sacrificed, and 2) the scoring system can represent a new standard in recording and reporting neonatal mouse health which has implications beyond this specific model. Given the tremendous number of biological differences between neonates and adults which have been revealed in the last few decades [4,19–24], it is no longer tenable to assume data generated from adult animal studies would be relevant to neonatal infectious disease. In order to study neonatal diseases, it is critical to work with neonatal animals, though this presents a unique set of challenges which may act as a barrier towards investigators who may consider working with neonatal mice[25]. The inability to ethically collect a meaningful volume of blood (>2 μL) without sacrifice precludes investigators from pairing survival data directly with data generated from any biological sample. Thus, the significance of this work is three-fold: (1) investigators may use this classification method to include mortality as a covariate of interest when all animals are sacrificed, (2) the scoring system can represent a new standard in recording and reporting neonatal mouse health which can standardize neonatal mouse work across different laboratories, and (3) animal suffering for future work can be minimized with a uniform, quantitatively established humane endpoint.

We also demonstrated that survival can still be assigned even for neonatal mice sacrificed 24 HPC; without such an approach, one is forced to rely on biomarkers (i.e. inflammatory cytokines, cell mobilization, etc.) as a proxy outcome. The decision to separate scores into their base components (righting reflex / mobility on each side) was made to maximize the amount of information being input into the classifier. Where human interpretation of data requires simplicity, i.e. a single score, machine learning approaches have the capacity to parse much more complex data and identify meaningful relationships. Given that it is increasingly unclear whether death from sepsis is associated with an inflammatory or an anti-inflammatory response [3,4,19,26,27], it is critical to minimize reliance on biomarkers, as most lack rigorous evidence of their relevance to survival. Our choice of bacterial load as an outcome of interest served a dual purpose, it: 1) validated the legitimacy of the scoring system and classifier, as a correlation between outcome and bacterial load is often assumed [11,28], and 2) provided evidence that there truly is a link between bacterial load and outcome in this model of polymicrobial sepsis. Thus, already at this stage we were able to remove one assumption which underlies neonatal mouse work. While a link between outcome and bacterial load may seem unsurprising, this demonstrates a proof of concept for situations wherein one must rely on a less obvious biomarker as a proxy outcome (i.e. sterile inflammatory models, inactivated bacteria, etc.). These health scores and classification approach provide a direct mechanism for assigning survival and non-survival rather than solely observing inflammatory markers and assuming a relationship which may or may not exist [16].

We were surprised to see that righting reflex alone was strongly correlated with bacterial load across all measured biological compartments, indicating it was a robust metric of neonatal mouse health (Fig 1). The significance of righting reflex was also apparent in the process of feature selection during classifier construction – righting reflex was the single most correlated feature with survival (Fig S4). Given the simplicity of collecting, these data suggest righting reflex should be recorded whenever neonatal mouse well-being is in question. Righting reflex in neonatal mice has previously been used as a tool to study behavioral development but has never been used to measure disease progression nor assess overall health in the context of an infection [29]. These data represent an important contribution towards describing neonatal mouse health which can help to inform ethical decisions around humane endpoints.

A potential limitation of this study is that mice which were non-mobile and failed to right on both sides (score 0) were sacrificed for ethical concerns, potentially violating the assumption of independence between score and survival. We addressed this by excluding mice who received a score of 0 from statistical analyses when survival was the outcome of interest. Further, the high bacterial load observed in whole-blood and the various organs independently validated the biological relevance of the scoring system presented. The use of righting reflex as a decision point in humane endpoint can help to standardize mouse models and reduce unnecessary suffering for newborn mice. In fact, the extremely low survival rate (<10%) of mice which failed to right and were lethargic 24 HPC suggests that this should be used as the humane endpoint for future studies. This approach could be further improved by rigorously selecting the times at which monitoring data were collected in order to maximize the potential difference between survivors and non-survivors whilst minimizing the potential cost of survivor bias. Finally, there are also some questions surrounding the use of righting reflex as a metric in younger mice which may be unable to right even at a healthy state – further research is warranted prior to assigning these health scores to mice prior to DOL 7.

The availability of a robust scoring system, with the additional proof that it can be predictive of outcome, should allow for a stratification of experimental groups beyond treated / untreated, or challenged / unchallenged, but also treated survivors / treated non-survivors, etc. The latter comparison will provide mechanistic insight into responders and non-responders of novel treatments – precisely what one hopes to attain from working with an animal model. Neonatal mouse models provide a mechanism to explore neonatal infectious disease in a manner that is unparalleled in humans. The ability to sacrifice and examine neonatal mice while not losing the potential associations with survival has the potential to break open the mysteries of neonatal sepsis. Here we demonstrated that bacterial load across all measured biological compartments was strongly correlated with our predictive health scores and therefore final outcome in neonatal polymicrobial sepsis. This study represents a blueprint for how to report neonatal mouse health in infectious disease models, and how to use that data to examine survival in mice which had to be sacrificed for sample collection.

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Supporting information captions

S1 Figure. Mortality and decrease in score were not a result of damage associated with injection. One mouse per litter received a sham challenge of either PBS or dextrose 5% water and the scores were recorded at 18 and 24 HPC. (A) Mice which received sham challenges exhibited no mortality. (B) Mice which received sham challenges had significantly higher scores at 18 HPC than their littermates (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$). In this cohort, most mice have recovered by 24 HPC so there is no significant difference between the groups ($p = 0.06$) but the sham challenged were clearly healthier.

S2 Figure. Distribution of scores assigned to neonatal mice at 18 and 24 hours post challenge. Scores assigned to neonatal mice 18 and 24 hours post IP challenge with cecal slurry (HPC). At 18 HPC the scores are poorly distributed, with the vast majority of mice assigned a score of 2 (failure to right, mobile) indicating that most mice have not progressed towards survival or non-survival. By 24 HPC the scores are evenly distributed as mice have begun to succumb to or recover from sepsis.

S3 Figure. Feature selection visualization using Pearson correlation. Heatmap of feature correlations of with Pearson correlation. Scores were split into components, starting with looking at righting reflex and mobility independent from one another and then further separated by the lower and higher measurements of each score taken in duplicate. Change in righting reflex reflects the difference between the monitoring timepoints at 18 HPC and 24 HPC.

S4 Figure. Evaluation of baseline and ensemble algorithms in their ability to classify survivors and non-survivors in a neonatal mouse model of polymicrobial sepsis. Algorithms were trained on a set of 148 pups and tested on another set of 74, the accuracy (number of correct classifications over total number of classifications made) is shown on the y-axis.

S5 Figure. 10-Fold cross validation of the Gradient Boosted Machine algorithm.

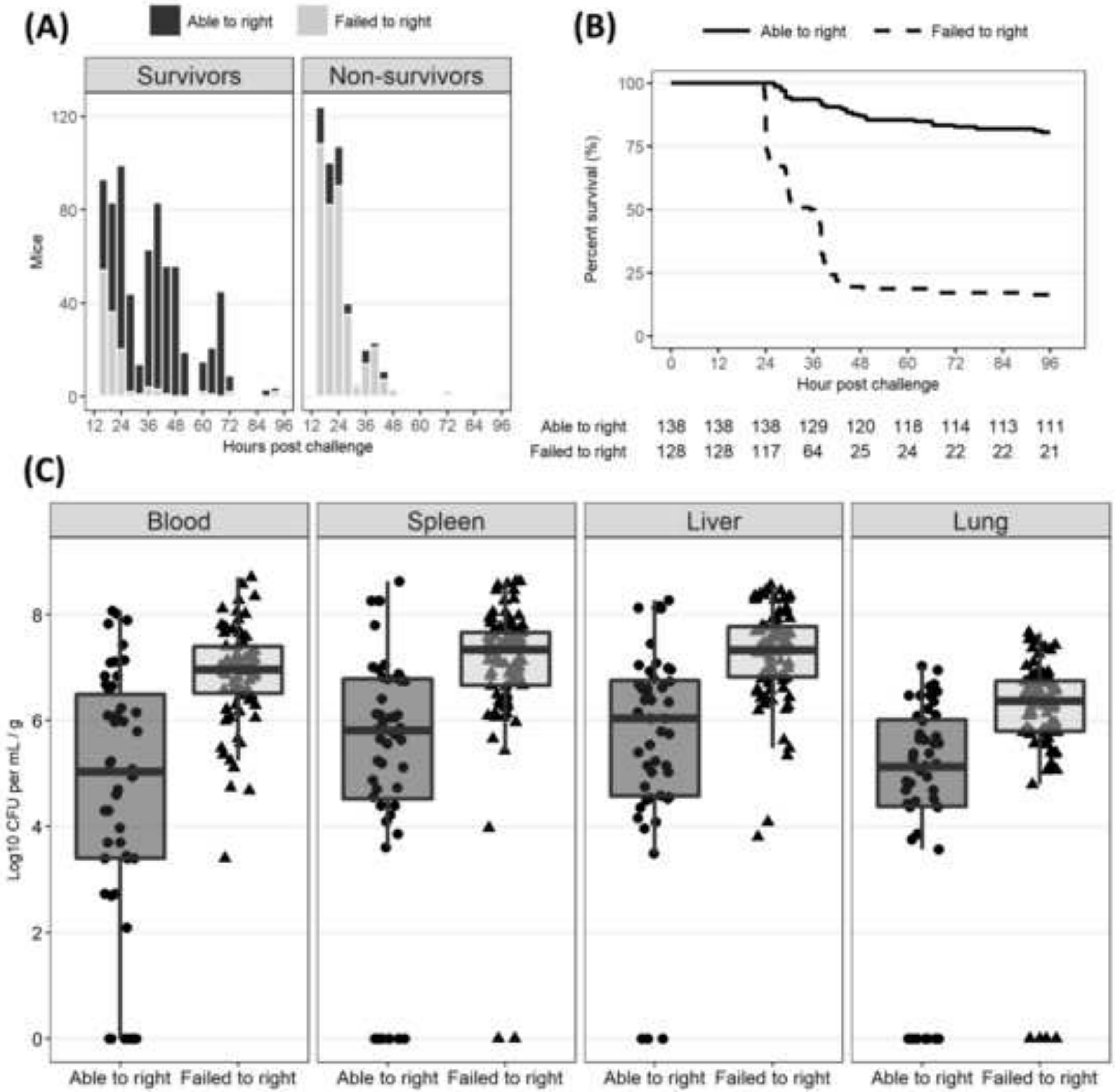
S1 Table. Accuracy of different algorithms and feature selection approaches in classifying mice into survivors or non-survivors at 24 HPC.

S2 Table. Confusion matrix showing Gradient Boosting Machine algorithm applied to external dataset. Confusion matrix demonstrating the predictive capacity of the classifier when applied to a never seen dataset of 21 neonatal mice challenged with the cecal slurry model of sepsis. The classifier accurately categorized 9/11 survivors and 9/10 non-survivors, in line with the internal cross-validation set.

S3 Table. User-defined hyperparameters to be tuned for each algorithm. Hyperparameters must be set prior to initiating the training process and are critical to constructing a robust classifier.

S1 URL. Public GitHub repository containing the code used to construct the classifiers. https://github.com/radaniba/Sepsis_Project

Figure 1


(A)
Able to right
Failed to right
Survivors
Non-survivors
Mice
Hours post challenge
(B)
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Percent survival (%)
Hour post challenge
Able to right 138 138 138 129 120 118 114 113 111
Failed to right 128 128 117 64 25 24 22 22 21
(C)
Blood
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Lung
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Able to right
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Figure 2

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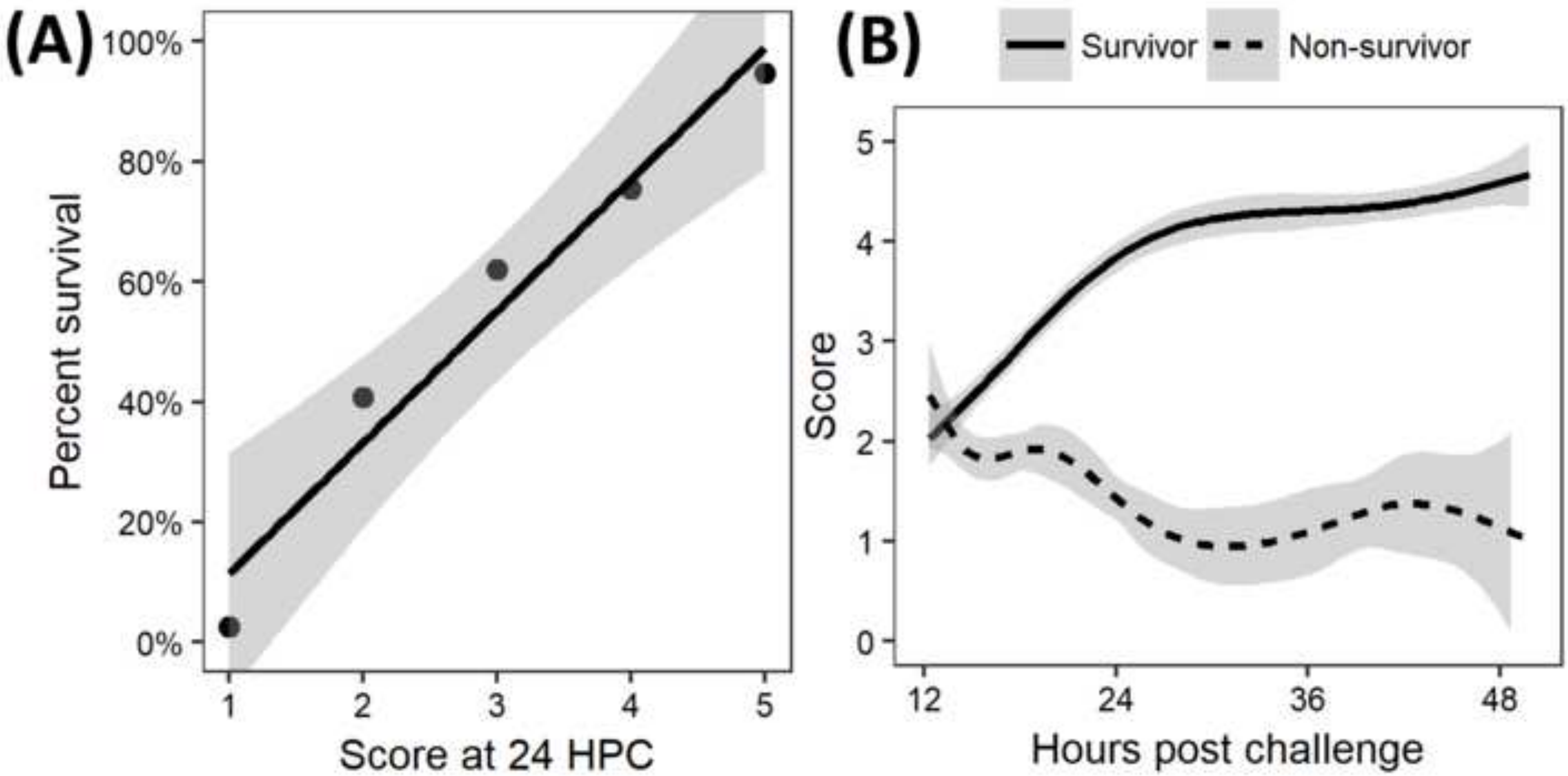

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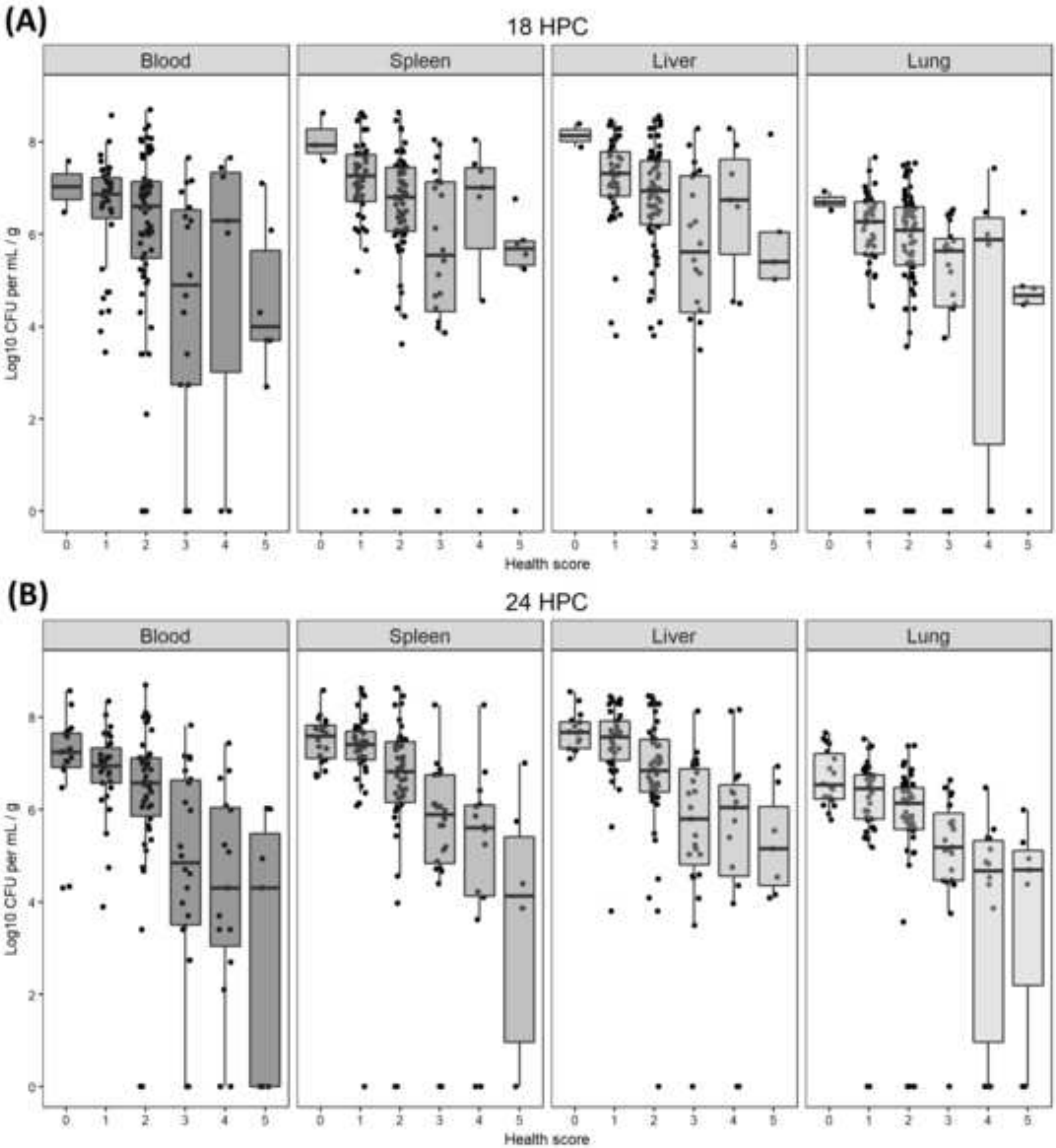

Figure 4

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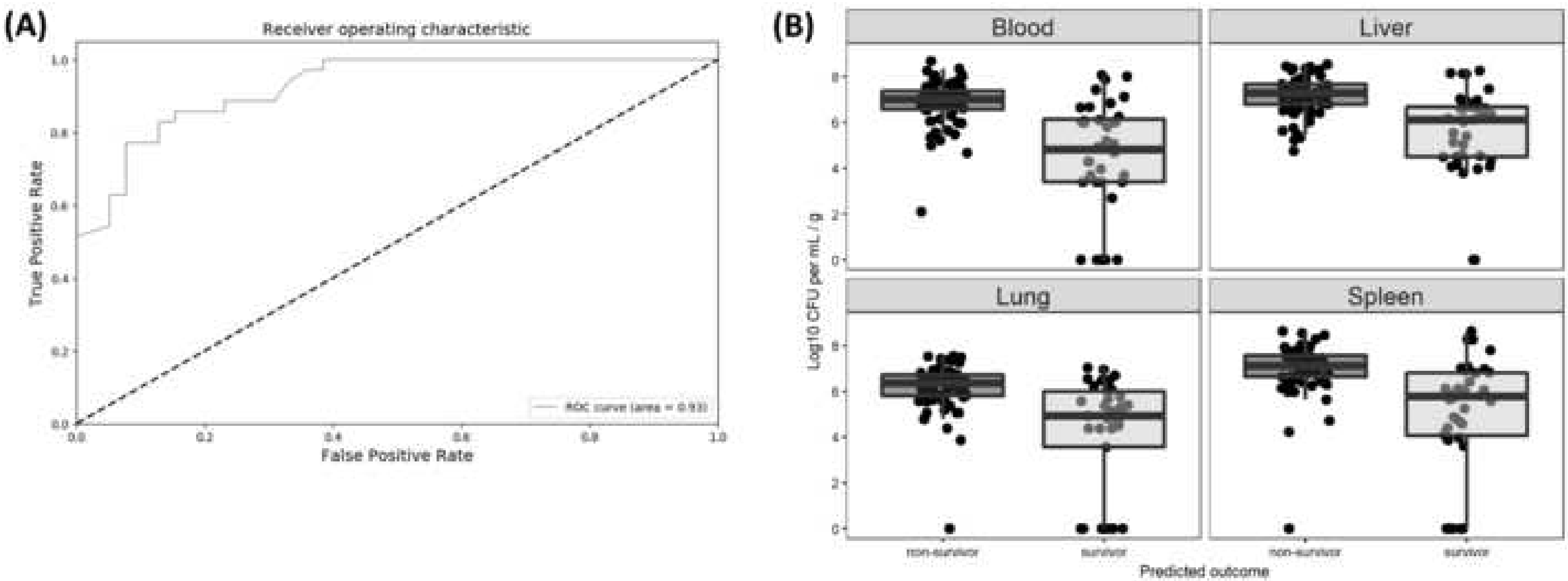

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Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis

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Abstract

Infectious disease and sepsis represent a serious problem for all, but especially in early life. Much of the ~~associated~~ increase in morbidity and mortality due to infection in early life is presumed to relate to fundamental differences between neonatal and adult immunity. Mechanistic insight into the way newborns' immune systems handle infectious threats is lacking; as a result, there has only been limited success in providing effective immunomodulatory interventions to reduce infectious mortality. Given the complexity of the host-pathogen interactions, neonatal mouse models can offer potential avenues providing valuable data. However, the small size of neonatal mice hampers the ability to collect biological samples without sacrificing the animals. Further, the lack of a standardized metric to quantify newborn mouse health increases reliance on correlative biomarkers without a known relationship to 'clinical' outcome. To address this bottleneck, we developed a system that allows assessment of neonatal mouse health in a readily standardized and quantifiable manner. The resulting health scores require no special equipment or sample collection and can be assigned in less than 20 seconds. Importantly, the health scores are highly predictive of survival. A classifier ~~constructed from the righting reflex and mobility~~built on our health score revealed a positive relationship between reduced bacterial load and survival, demonstrating how this scoring system can be used to bridge the gap between assumed relevance of biomarkers and the clinical outcome of interest. Adoption of this scoring system will not only provide a robust metric to assess health of newborn mice but will also allow for objective, prospective studies of infectious disease and possible interventions in early life.

Introduction

An estimated 3.0 million cases of newborn infection every year result in nearly 350,000 deaths, making infection one of the most lethal threats in early life [1]. Many deaths due to infection can be attributed to sepsis, a difficult to define disease characterized by failures of regulatory immune system components resulting in hyper- or hypo-inflammation [2,3]. It is widely known that the neonatal immune system is fundamentally distinct from that of adults, indicating that data generated from adult animal models should not be assumed to apply to newborns [4–7]. While neonatal mouse models offer several practical benefits (availability, existing infrastructure, well-understood biology, etc.), working with neonatal mice presents some unique challenges which have not yet been effectively addressed. In particular, researchers are limited by the size of the mice; ~~near~~ day of life (DOL) 7 mice (considered to be immunologically closest to human term newborns at birth) [8] ~~mice~~ are still far too small to collect meaningful volumes of blood without sacrificing experimental animals. In adult mice, about 7 to 8 µL of blood can ethically be collected per gram of mouse bodyweight without sacrifice or fluid replacement – this translates to a maximum single collectible blood volume from neonatal mice of about 20 µL [9]. Serial sampling (every 24 hours) would only allow for collection of 5 µL of blood or less in each draw [10]. This limitation (in addition to the practical difficulties of drawing blood from such small animals) only allows terminal methods of bleeds to collect samples of adequate volume~~leads to sample collection at time of sacrifice~~, which fundamentally separates biological findings from true 'clinical' outcomes (i.e. survival). This inability to serially sample neonatal mice without sacrifice also forces researchers to rely heavily on biomarkers as a stand-in for outcome when exploring a potential treatment of interest, assuming causality between e.g. pathogen load or inflammatory cytokine production and mortality.

Health scores for sepsis are widely used in human clinical settings as they are critical for capturing disease progression and can inform potential interventions [11–15]. The ability to assess disease severity in a quantitative way is also extremely useful in animal models as it can help investigators avoid an overreliance on assumptions about the relevance of a given biomarker. For example, one group developed a series of symptom scores for adult mice in a similar model of sepsis (fecal suspension injected IP) which relied on, among other things: activity, response to stimulus, eyes (open or shut), level of consciousness, and appearance [16]. This study illustrated that when grouping adult mice by these health scores, TNFα and IL-1β levels did not correlate with outcome despite elevated levels in comparison to unchallenged controls – these cytokines went up following challenge but did not appear different in survivors and non-survivors. This demonstrates the capacity for health scores to narrow the focus of investigators onto biomarkers which are functionally relevant to a clinical outcome of interest – a capacity even more critical when working in a neonatal population from which even blood samples are extremely difficult to collect without harming them. While the quantitative observations used in this model may be strong ~~markers~~ markers for adult animals, none of these characteristics are readily applicable to a neonatal population (i.e. no fur to be ruffled, eyes are always closed, minimal basal activity levels, etc.).

~~We~~ To address this need, we developed a data-driven and simple scoring system ~~that is~~ directly related to outcome (humane endpoint) in neonatal mice. The addition of our health scores to neonatal mouse experiments provides: (a) an improved ability to track~~ing of~~ disease progression which can inform time of sacrifice for sample collection and (b) a quantitative method to prospectively differentiate survivors

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from non-survivors. We here ~~validated~~ correlate this ~~six-point~~ health scoring system ~~(0-5)~~ with mortality in a polymicrobial model of neonatal sepsis. This model confirms that clinical outcome ~~is~~ was highly correlated with bacterial load. It ~~takes~~ took less than 20 seconds to determine these composite health scores per animal. The prospective validity of our health score in assigning outcome provides the much-needed metric to begin assessing causality of biomarkers, which in turn will allow assignment of mechanistic cause-effect relationships. This approach will not only decrease undue suffering of the animals, but also increases the likelihood of identifying effective immunomodulatory interventions for infectious disease and sepsis.

Materials and Methods

Mice

Male and female C57BL/6 mice aged 6 weeks (Jackson Labs, catalogue #664) were purchased from The Jackson Laboratory and used for breeding purposes; all mice used in this study were first or second-generation progeny of Jackson mice bred in-house. This study was carried out in strict accordance with animal ethics guidelines outlined by University of British Columbia and was approved by the Animal Care Committee (protocol numbers A17-0110 and A14-0261). Animals were housed in OptiMice cages, with food and water *ad libitum*. Breeding mice were fed a high fat diet (Tekland Cat 2919) while others were fed a low-fat diet (Tekland Cat 2918). The pellet base was cellulose ¼" Performance Bedding (Lab Animal Supplies INC, Cat L0108), and environmental enrichment included a red hut, nesting material of Envirodri (Sic.) (Jameson, B501) and Nestlets (Ancare, Cat CABFM00088). Mice underwent a controlled 12 hour light cycle, with temperature maintained at 23 °C. Mice were anesthetized with isoflurane prior to euthanasia through CO_2 exposure, confirmed by decapitation. Humane endpoint was defined as the lack of an attempt to right themselves when placed on their back on both sides, explained in greater detail below.

Cecal slurry model of neonatal sepsis

Polymicrobial sepsis was induced using a modified version of the model first described by Wynn *et al.* in 2007 and recently outlined on video in the Journal of Visualized Experiments [17,18]. Briefly, cecal material was extruded from adult male mouse ceca (aged 6-10 weeks), diluted in dextrose 5% water (D5W) to 160 mg slurry / mL D5W, filtered through a 70 µm sterile strainer, pooled, aliquoted, and frozen at -80 ^{o}C. Aliquots were thawed at room temperature, diluted further in D5W to the desired weight-adjusted dose, and kept on ice prior to intraperitoneal (IP) injection. The challenge dose was ~~titered~~ titrated to achieve a lethal dose equal to 50% (LD50) per litter. Neonatal mice were injected IP with approximately 100 µL cecal slurry (0.8 – 1.1 mg/g mouse) on ~~DOL~~ day of life (DOL) 7 or 8 to induce polymicrobial sepsis, with slight variations in volume due to weight adjustment. Litters were monitored for morbidity (i.e. ~~humane endpoint~~lethargy, inability to right) up to 96 hours post challenge (HPC). In these survival experiments, mice were monitored either until full recovery was observed (gaining weight, alert / active), or until humane endpoint was reached at which point the mice would be euthanized. A separate cohort of ~~mice~~ pups were sacrificed at approximately 24 HPC in order for the collection of blood and tissue samples. ~~with bacterial~~ Bacterial loads in liver, lungs, spleen, and blood were subsequently calculated by serially diluting the organ homogenates in PBS, ~~and~~ drop-plating on 5% sheep's blood agar, and the counting colony forming units (CFU) after 24 hours of incubation at 37 ^{o}C.

~~Mice~~

~~Male and female C57BL/6 mice aged 6 weeks (Jackson Labs, catalogue #664) were purchased from The Jackson Laboratory and used for breeding purposes; all mice used in this study were first or second-~~

~~generation progeny of Jackson mice bred in-house. This study was carried out in strict accordance with animal ethics guidelines outlined by University of British Columbia and was approved by the Animal Care Committee (protocol numbers A17-0110 and A14-0261). All efforts were taken to minimize animal suffering. Mice were anesthetized with isoflurane prior to euthanasia through CO_2 exposure, confirmed by decapitation.~~

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Monitoring and health scores

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Monitoring commenced 12 HPC (i.e. prior to the onset of expected mortality) and continued 2-3 times daily (07:00 – 09:00, 12:00-14:00, 16:00-18:00, spanning both light and dark cycles) through to experimental endpoint. All mice were challenged in the evening (17:00 – 18:00) so monitoring schedules were not adjusted based on time of challenge. During mouse scoring the neonatal mice were separated into a secondary cage to avoid ~~dam~~ stressing the dam. Nesting material was rubbed in hands to transfer that litter's smell to the gloves, and then each mouse was individually scruffed and placed on their back as visualized in Brook *et al.* 2019 [18]. Health scores arose naturally out of a need to define a humane endpoint for extremely ill neonatal pups. Placing a pup on its back and recording whether it was able to right itself was a simple and effective discriminator of outcome, but alone it failed to capture the obvious difference between mice which were vigorously attempting to right and those which barely moved. Thus, we began recording qualitative observations of vigor to pair with righting reflex data, which eventually became quantitative descriptors of mobility (i.e. angle of hip movement when attempting to right). Righting reflex measurements were further standardized by adding a four-second cutoff, determined by the observation that more time was unnecessary, and less time risked inappropriately assigning 'fail to right' to a healthy pup. For simplicity of recording and interpretation, these observations were transformed into ranked scores from 0 - 5, as presented in Table 1. Scores were recorded at each monitoring time point in duplicate. In addition to scoring, mice were weighed at each time point using a top-loading balance. All mice used were either 7 or 8 days old and weighed an average of 3.7 g (2.25 g - 6.06 g). Pups were identified by markings with an ethanol proof pen on the top or bottom of the tail. Monitoring sheets for each litter were used to record scores and weights. ~~Mice analyzed for the present study were principally untreated controls from other experiments not shown here, hence the large sample size (404~~ A total of 424 mice ~~total~~ were used in this study: 266 with an endpoint of mortality or recovery, 125 sacrificed at 24 hours for a readout of bacterial load, 33 used in follow-up experiments~~)~~. In survival experiments, all mice were allowed to proceed to humane endpoint or recovery (gaining weight four days post-challenge with scores of 4-5), whereas in bacterial load experiments all mice were sacrificed at 24 HPC regardless of perceived illness.

~~Humane endpoint was defined as the lack of an attempt to right themselves when placed on their back on both sides ("failure to right; non-mobile", further explained in main text). Monitoring commenced 12 HPC (i.e. prior to the onset of expected mortality) and continued 2-3 times daily through to experimental endpoint. All mice used were either 7 or 8 days old and weighed an average of 3.7 g (2.25 g - 6.06 g).~~

Classifier

The ability to confidently predict outcome (survival or non-survival) at time of sacrifice would greatly help limit the number of assumptions required by investigators ~~regarding the relevance of biomarkers and acted as evidence of the utility of our newly developed scoring system~~. We ~~built a~~ interpreted survival vs. ~~or~~ non-survival as a binary classification problem and constructed multiple machine learning models aimed at classifying mice into one of these two groups at 24 HPC. ~~for a binary classification problem, assigning mice to either "survivor" or "non-survivor" categories.~~ In order to identify the best model, we tested our dataset in 6 different algorithms~~, including~~ split across two primary classes: baseline learning algorithms (Logistic Regression, K Nearest Neighbors and Decision Tree) ~~as well as~~ and ensemble learning algorithms (Random Forest, Gradient Boosting and XGBoost). Feature selection began from all available data collected for each mouse: sex, date of birth, time of challenge, weight at challenge, weight at each monitoring time point, precise HPC each monitoring time occurred, righting reflex, mobility at each monitoring time point, and the change in any of the above from one point to another. ~~Various approaches were tested for feature selection, but the final classifier relied on~~ Pearson correlation coefficients were used to avoid including highly co-correlated and irrelevant features, with a correlation threshold of 0.2 (with outcome) acting as the minimum cutoff for inclusion. This threshold and all other parameters were optimized iteratively using GridSearch, which examines all possible combinations of values for each parameter and tunes each parameter to maximize classification strength. A~~ a~~ more detailed description of the other approaches we examined for feature selection is presented in the supplementary material (S1 Appendix). ~~Feature selection and classification was performed on components of the scores (righting reflex and mobility) rather than directly on the scores themselves in order to independently quantify the significance of each metric, in comparison to other commonly collected biometric data (i.e. weight).~~ For specific parameters used during classifier construction, see the source code linked in the supplementary materials (S1 URL). The righting reflex and mobility were treated separately (rather than combined as scores) to maximize the amount of information available for the classification problem and minimize the potential for biasing what should be an unsupervised process with our own assumptions about what should be associated with mortality. A total of 222 mice were split into a training set (148 mice) and a test set (74 mice) and the different approaches were evaluated by accuracy (total number of correct classifications over total number of instances) and area under the receiver operating characteristic curve (AUC), which quantified sensitivity and specificity and represented a good metric of the quality of a classifier (S1 Table). Evaluation based on these performance metrics led us to select Gradient Boosted ~~machine~~ Machine learning ~~was selected~~ as the strongest classification ~~algorithm~~algorithm and was applied to an external data set of 21 mice for further validation. ~~based on accuracy and AUC (Table S1) and was applied to an external dataset for further validation. For a~~ A ~~more~~ detailed description of classifier construction and background behind the various approaches examined~~,~~ ~~see~~ are available in the supplementary material (S1 Appendix).

Statistics

Comparisons of survival curves were evaluated using the log-rank test, with statistical significance assigned when $p < 0.05$. All tests comparing score and survival excluded the assigned score of 0, as this was the defined humane endpoint and thus could not be treated as independent. Percent mortality was analyzed using two-sided Fischer's exact tests with Bonferroni adjustment where appropriate. Scores were only treated as numerical quantities for the purpose of examining change over time – all other ~~instances~~analyses treated scores as categorical. Any statistical test applied identically to blood, spleen, liver, and lung was adjusted using the Bonferroni method. Bacterial load data did not exhibit a normal distribution (per Shapiro-Wilk normality test) and were compared using the Wilcoxon rank-sum test. When analyzing independent variables with multiple levels, data were first tested with the non-parametric Kruskal-Wallis test and only ~~upon achieving~~if significant~~ce~~ ($p < 0.05$) were post-hoc, pairwise comparisons made using the Wilcoxon rank-sum test (Benjamani-Hochberg adjusted). All statistical analyses were performed in R (version 3.5.4).

Results

~~Creating health scores~~Health scores and outcome

At all monitoring time points, righting was associated with survival and failure to right (FTR) was associated with non-survival (Fig 1A). As 24 HPC was as far into disease progression as one could wait to sacrifice mice without incurring heavy survivor bias (Fig ~~1b~~1B), this timepoint became the time of sacrifice for the alternative cohort of mice assessed for bacterial load. It was therefore critical to be able to distinguish survivors and non-survivors prior to sacrifice and~~t~~ no later than 24 HPC. Mice which failed to right at 24 HPC were significantly more likely to die than those which successfully righted (log-rank test, p < 0.001) and had a statistically significant, 100-fold higher bacterial load across all compartments (~~independent two-group t-tests~~two-sided Wilcoxon rank-sum tests with Bonferroni correction, p < 0.001) (Fig ~~1c~~1C). Righting reflex alone was strongly correlated with survival and bacterial load, especially after 24 HPC. We also captured three different levels of mobility: non-mobile, lethargic, and ~~active~~ mobile (Table 1). ~~-~~One of the strengths of a scoring system should be the ability to provide a metric predictive of future outcome. To assess the predictive value of this righting / mobility combination metric, we assigned ~~numerical~~ scores from 0 (fail to right non-mobile, humane endpoint) to 5 (rights mobile, perfectly healthy) (Table 1). Unless otherwise indicated, the reported score is the lower of the two recorded for each mouse.

Figure 1. Righting reflex at 24 HPC is an excellent discriminator of survival and bacterial load. Righting reflex of neonatal mice (DOL 7-8) challenged IP with cecal slurry at an LD50~~(0.8 -1.1 mg/g)~~. (A) Ability to right at different hours post challenge (HPC) in survivors and non-survivors. (B) Survival curve of mice split by righting reflex at 24 HPC. Mice which were able to right themselves at 24 hours were significantly more likely to survive the cecal slurry challenge than mice which failed to right (log-rank test, p < 0.001). (C) Bacterial load split by righting reflex at time of sacrifice, 24 HPC. Mice which were able to right had~~a~~ significantly lower bacterial load in ~~the whole~~ blood and all measured organ tissues (two-sided Wilcoxon rank-sum tests ~~independent two-group t-tests~~ with Bonferroni correction, p < 0.001).

Table 1. Assigning numerical health scores to quantitative observations in neonatal mice

Numerical Score	Righting Reflex*	Mobility	Description
5	Rights	Mobile	• Multiple steps in a row, responsive to tail tap • Alert, may be shaky but moving with energy • Might fall over while walking
4	Rights	Lethargic	• Takes at least one step, but stops between steps • Steps are slower, may be very shaky, may fall over
3	Rights	Nonmobile	• Stationary after righting • May take a half-step after tail tap but largely or entirely unresponsive
2	Fails to right	Mobile	• Vigorous hip movement, swinging side to side at an angle ≥ 90° from horizontal at least once during the observation period (4 seconds)
1	Fails to right	Lethargic	• Slower hip movement, ≤ 90° angle • Halting attempts to right over the observation period, may be some pauses but always starting again
0	Fails to right	Nonmobile	• Zero or extremely weak hip movement • Gives up attempting to right or never starts, may be uncontrollably shaky • limbs may bend at the knee and extend but hips will not rock side to side

*Righting reflex measured by placing mice gently on their back and observing their ability to right within 4 seconds. All scores were taken in duplicate with the lower of the two reported here.

Health scores are strongly associated with survival and bacterial load

Combining mobility with righting reflex allowed us to generate six categories which were assigned to numerical scores for ease of recording and visualizing. As the score of zero was not independent from mortality (score of zero was the defined humane endpoint and necessitated euthanasia), it was excluded from statistical analyses. Mice which received sham challenges with either phosphate buffered saline (PBS) or dextrose 5% water (D5W) exhibited no mortality and had significantly higher scores than those which received cecal slurry, indicating that the drop in score post-challenge was not a reflection of injury due to the injection itself but rather was in response to disease (S1 Fig). Comparing the categorical scores at 18 and 24 HPC indicated a significant relationship between score and survival (Fisher's exact test, p < 0.001). A simple linear regression model examining percent overall survival against assigned score at 24 HPC (mice with a score of 0 were excluded) was found to be significant ($R^2 = 0.946$, p = 0.0035, n = 210) with each increase in score equating to a 22% ± 8% improved chance of survival (Fig 2A). Visualizing the changes in score over the first 48 hours of disease (capturing 93% of deaths) showed a clear separation between survivors and non-survivors beginning around 18 HPC, with higher scores associated with recovery and survival (Fig 2B).

Figure 2. Health scores ~~are~~ were directly associated with outcome in a polymicrobial model of sepsis in neonatal mice.

(A) Scores recorded at 24 hours post challenge (HPC) plotted against the percent of mice which survived the IP cecal slurry challenge. Scores of zero were excluded as they were the pre-determined humane endpoint. A significant linear regression was found ($R^2 = 0.946$, p = 0.0035, n = 210) with percent survival increasing by 22 ± 8% for each single point increase in score. Shaded area represents standard error. (B) Visual representation of change in score defined numerically over time following IP cecal slurry challenge.

The inability to distinguish survivors through health scores at 12 HPC (Fig 2B) was consistent with qualitative observations that most mice appeared similar during this time period irrespective of final outcome. At 24 HPC, the scores were well distributed with the least frequent score (3) assigned to 10.9% of mice (29/266) and the most frequent, the healthiest score of 5, assigned to 21.0% (56/266) of mice (S2 Fig ~~S1~~). At 18 HPC, many of the mice appeared similarly ill, with only 6.9% at the healthiest score of 5 and 46.8% at the middle score of 2. Thus, health scores at 24 HPC ~~are~~were most useful in their ability to capture a range of sickness in neonatal mice.

~~**Figure S1. Frequency of scores observed at 18 and 24 hours post challenge.**~~

~~Assigned health score distribution 18 and 24 hours after IP cecal slurry challenge of neonatal mice. Disease progression towards resolution or humane endpoint was demonstrated by the relative increase in scores on either extreme (0 and 5).~~

To test the validity of our scoring system, a separate cohort of 125 mice generated under the same experimental conditions were sacrificed at 24 HPC and samples of whole-blood, spleen, liver, and lung tissue were plated on sheep's blood agar and the colony forming units (CFU) growth was assessed 24 hours later. Two separate~~, one-way multivariate analyses of variance (MANOVAs)~~, non-parametric Kruskal-Wallis tests were performed to assess the relationship between categorical scores at 18 and 24 HPC and bacterial load (at 24 HPC) across all four biological samples. A significant ~~MANOVA~~ effect was attained both at 18 HPC (~~Pillais' Trace = 0.24, *F*(20, 572) = 1.85,~~ Bonferroni-adjusted p-values ~~=~~< 0.001~~276~~) and 24 HPC (~~Pillais' Trace = 0.44, *F*(20, 5.72) = 3.50,~~ Bonferroni-adjusted p-values < 0.001) (Fig 3). The distribution of scores broadened as disease progressed towards an outcome – nearly all mice scored between 1-3 at 12 HPC, most fell within that same range at 18 HPC, but an even distribution was present at 24 HPC (S2 Fig ~~S1~~). Our health scores were strongly associated with both survival and bacterial load at 24 HPC.

Figure 3. Health scores at 18 and 24 HPC are related to bacterial load at 24 HPC.

(A) A Kruskal-Wallis comparison of score 18 hours post challenge (HPC) with bacterial load in whole-blood and tissues showed a significant ~~MANOVA~~ effect (~~Pillais' Trace = 0.24, *F*(20, 572) = 1.85,~~ Bonferroni-adjusted p-values < 0.001~~ = 0.0276~~) indicating score at 18 HPC ~~is~~ was related to the bacterial load measured in tissues of the same animals collected 6 hours later (24 HPC). (B) A significant ~~MANOVA~~

effect was also attained at 24 HPC (~~Pillais' Trace = 0.44, *F*(20, 5.72) = 3.50,~~ Bonferroni-adjusted p-values < 0.001) demonstrating the relationship between health scores and bacterial load across all biological compartments.

Righting reflex and mobility are predictive of outcome prior to sacrifice

We used a machine learning approach to classify mice as survivors or non-survivors based primarily on righting reflex, mobility, and weight, specifically looking at how each metric changed from 18 to 24 HPC. Only 12% (28 / 222) of mice were at humane endpoint at 24 HPC, so while the association between score components and outcome may be influenced by the inclusion of these mice (mice which were FTR non-mobile on both sides are at humane endpoint and must be sacrificed), they did not represent a large enough proportion to call the results into question. Pearson correlation proved the most effective method of feature selection and identified attributes most associated with death: a mobility score of "non-mobile", a change from able to right at 18 HPC to failed to right at 24 HPC, and weight loss between 18 and 24 HPC (Fig S4). The ability to right was strongly correlated with survival, especially the consistent ability to right at both 18 and 24 HPC.

We used 10-fold cross validation training on a combination of six algorithms and three methods of feature selection in order to identify the most effective method of classification (~~Table~~ S1 Table). A more detailed description of this approach is available in the supplemental section appendix 1 (S1 Appendix). For the final predictions, we used Gradient Boosting Machine as our classifier, combined with features selected by Pearson correlation coefficients. This approach had a cross validation average accuracy of 0.91 with a low standard deviation of 0.06, indicating little risk of over or underfitting (S3 Fig ~~S2~~). With

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the hyperparameters selected for our Gradient Boosting model, we built a model on the training data subset with known outcome and applied it to 33% of the original data set not used during the training phase. With a total of 74 cases to classify as “survivor” or “non-survivor”, the Gradient Boosting model performed well with an accuracy score of 0.85 and an AUC of 0.93 (Fig 4A), with 33 true positive cases classified as “non-survivor” and 30 cases classified as “survivor” (~~Fig 4~~Table 2). To further validate our model, we generated an additional dataset with the same data collection process as well as the same data cleaning and standardization pipeline. With a total of 21 new data points, our model ~~was able to~~ accurately classif~~y~~ied 18 with an average score of 85% accuracy (~~Fig S3~~S2 Table). Righting reflex and mobility at 24 hours together with change in weight were the features most strongly correlated with outcome (~~Fig S4~~S3 Figure). Finally, we applied our classifier to ~~the~~ a~~subset~~ a different set of ~~mice~~ pups ~~which were~~ sacrificed at 24 HPC and identified a clear relationship between survival and bacterial load (Fig ~~5~~4B).

Table 2. Confusion matrix showing accuracy of Gradient Boosting Machine model when applied to test set of 74 mice. Using monitoring data collected at 18 and 24 HPC (weight, righting reflex and mobility), a Gradient Boosted machine learning model was able to distinguish survivors and non-survivors with an accuracy score of 0.85; 33/39 survivors and 30/35 non-survivors were correctly classified.

True class	Survivor (Predicted class)	Non-Survivor (Predicted class)
Survivor	**33**	6
Non-survivor	5	**30**

Figure 4. ~~**Righting reflex, mobility, and biometric data can predict survival or non-survival prior to sacrifice in neonatal mice.**~~ **Survival or non-survival of neonatal mice was predicted at 24 HPC and was strongly associated with bacterial load.**

~~(A) Confusion matrix showing the accuracy of the classification model applied to the test dataset, the model is accurately able to distinguish the survivor class from the non-survivor class with an 85% accuracy.~~ (~~B~~A) Receiver Operating Characteristic ~~ROC~~ curve illustrating the sensitivity and specificity of ~~the model applied to the~~ a Gradient Boosting Machine model trained on 148 mice and tested here on 74 mice. The model was primarily driven by weight change, righting reflex and mobility at 18 and 24 HPC. (B) Neonatal mice injected IP with cecal slurry were sacrificed 24 hours post challenge, with tissue and blood samples plated on sheep's blood agar at time of sacrifice. Mice were predicted to survive or not survive based on righting reflex, mobility, and other biometric data. Mice which were predicted to survive had a significantly lower bacterial load in all measured biological compartments than predicted non-survivors (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$). All mice with zero bacterial load in the whole-blood (n=9) were predicted to survive.

~~test set.~~

~~**Figure 5. Bacterial load is strongly associated with predicted outcome.**~~

~~Neonatal mice injected IP with cecal slurry were sacrificed 24 hours post challenge, with tissue and blood samples plated on sheep's blood agar at time of sacrifice. Mice were predicted to survive or not survive based on righting reflex, mobility, and other biometric data. Mice which were predicted to survive had a significantly lower bacterial load in all measured biological compartments than predicted non-survivors (Independent two-group t-tests with Bonferroni correction, p < 0.001). All mice with zero bacterial load in the whole-blood (9) were predicted to survive.~~

Discussion

Here we present~~ed~~ a system of health scores for neonatal mice which were used to confidently predict mortality 24 HPC in an established model of neonatal sepsis [17]. The significance of this is twofold: 1) the classifier enables investigators to include “clinical” outcome (e.g. mortality) as a covariate of interest when all animals are sacrificed, and 2) the scoring system can represent a new standard in recording and reporting neonatal mouse health which has implications beyond this specific model. Given the tremendous number of biological differences between neonates and adults which have been revealed in the last few decades [4,19–24], it is ~~irresponsible~~no longer tenable to assume data generated from adult animal studies would~~ill~~ be relevant to neonatal infectious disease. In order to study neonatal diseases, it is critical to work with neonatal animals, though this presents a unique set of challenges which may act as a barrier towards investigators who may consider working with neonatal mice[25]. The inability to ethically collect a meaningful volume of blood (>2 µL) without sacrifice precludes investigators from pairing survival data directly with data generated from any biological sample. Thus, the significance of this work is three-fold: (1) investigators may use this classification method to include mortality as a covariate of interest when all animals are sacrificed, (2) the scoring system can represent a new standard in recording and reporting neonatal mouse health which can standardize neonatal mouse work across different laboratories, and (3) animal suffering for future work can be minimized with a uniform, quantitatively established humane endpoint.

We also demonstrated that survival can still be ~~assessed~~ assigned even for ~~in~~ neonatal mice ~~which were all~~ sacrificed 24 HPC; without such an approach, one is forced to rely on biomarkers (i.e. inflammatory cytokines, cell mobilization, etc.) as a proxy outcome. The decision to separate scores into their base components (righting reflex / mobility on each side) was made to maximize the amount of information being input into the classifier. Where human interpretation of data requires simplicity, i.e. a single score, machine learning approaches have the capacity to parse much more complex data and identify meaningful relationships. Given that it is increasingly unclear whether death from sepsis is associated with an inflammatory or an anti-inflammatory response [3,4,19,26,27], it is critical to minimize reliance on biomarkers, as most lack ~~without some~~ rigorous evidence of their relevance to survival. Our choice of bacterial load as an outcome of interest served a dual purpose, it: 1) validated the legitimacy of the scoring system and classifier, as a correlation between outcome and bacterial load is often assumed [11,28], and 2) provided evidence that there truly is a link between bacterial load and outcome in this model of polymicrobial sepsis. Thus, already at this stage we were able to ~~have~~ remove~~d~~ one assumption which underlies neonatal mouse work. While a link between outcome and bacterial load may seem unsurprising, this demonstrates a proof of concept for situations wherein one must rely on a less obvious biomarker as a proxy outcome (i.e. sterile inflammatory models, inactivated bacteria, etc.). These health scores and classification approach provide a direct mechanism for assigning survival and non-survival rather than solely observing inflammatory markers and assuming a relationship which may or may not exist [16].

We were surprised to see that righting reflex ~~in of itself~~alone was strongly correlated with bacterial load across all measured biological compartments, indicating it was a robust metric of neonatal mouse health (Fig 1). The significance of righting reflex was also apparent in the process of feature selection during classifier construction – righting reflex was the single most correlated feature with survival (Fig S4). Given the simplicity of collecting, these data suggest ~~it~~ righting reflex should be recorded whenever neonatal mouse well-being is in question. Righting reflex in neonatal mice has previously been used as a tool to study behavioral development but has never been used to measure disease progression nor assess overall health in the context of an infection [29]. These data represent an important contribution towards describing neonatal mouse health which can help to inform ethical decisions around humane endpoints.

A potential limitation of this study is that mice which were non-mobile and failed to right on both sides (score 0) were sacrificed for ethical concerns, potentially violating the assumption of independence between score and survival. We addressed this by excluding mice ~~at~~ who received a score of 0 from statistical analyses when survival was the outcome of interest. Further, the high bacterial load observed in whole-blood and the various organs independently validated the biological relevance of the scoring system presented. The use of righting reflex as a decision point in humane endpoint can help to standardize mouse models and reduce unnecessary suffering for newborn mice. In fact, the extremely low survival rate (<10%) of mice which failed to right and were lethargic 24 HPC suggests that this should be used as the humane endpoint for future studies. This approach could be further improved by rigorously selecting the times at which monitoring data were collected in order to maximize the potential difference between survivors and non-survivors whilst minimizing the potential cost of survivor bias. Finally, ~~There~~ there ~~is also~~ are also some questions surrounding the use of righting reflex as a metric in younger mice which may be unable to right even at a healthy state – further research is warranted prior to assigning these health scores to mice prior to DOL 7.

The availability of a robust scoring system, with the additional proof that it can be predictive of outcome, should allow for a stratification of experimental groups beyond treated / untreated, or challenged / unchallenged, but also treated survivors / treated non-survivors, etc. The latter comparison will provide mechanistic insight into responders and non-responders of novel treatments – precisely what one hopes to attain from working with an animal model. Neonatal mouse models provide a mechanism to explore neonatal infectious disease in a manner that is unparalleled in humans. The ability to sacrifice and examine neonatal mice while not losing the potential associations with survival has the potential to break open the mysteries of neonatal sepsis. Here we demonstrated that bacterial load across all measured biological compartments was strongly correlated with our predictive health scores and therefore final outcome in neonatal~~,~~ polymicrobial sepsis. This study represents a blueprint for how to report neonatal mouse health in infectious disease models, and how to use that data to examine survival in mice which had to be sacrificed for sample collection.

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Supporting information captions

S1 Figure. Mortality and decrease in score were not a result of damage associated with injection. One mouse per litter received a sham challenge of either PBS or dextrose 5% water and the scores were recorded at 18 and 24 HPC. (A) Mice which received sham challenges exhibited no mortality. (B) Mice which received sham challenges had significantly higher scores at 18 HPC than their littermates (two-sided Wilcoxon rank-sum tests with Bonferroni correction, $p < 0.001$). In this cohort, most mice have recovered by 24 HPC so there is no significant difference between the groups ($p = 0.06$) but the sham challenged were clearly healthier.

S2 Figure. Distribution of scores assigned to neonatal mice at 18 and 24 hours post challenge. Scores assigned to neonatal mice 18 and 24 hours post IP challenge with cecal slurry (HPC). At 18 HPC the scores are poorly distributed, with the vast majority of mice assigned a score of 2 (failure to right, mobile) indicating that most mice have not progressed towards survival or non-survival. By 24 HPC the scores are evenly distributed as mice have begun to succumb to or recover from sepsis.

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S3 Figure. Feature selection visualization using Pearson correlation. Heatmap of feature correlations of with Pearson correlation. Scores were split into components, starting with looking at righting reflex and mobility independent from one another and then further separated by the lower and higher measurements of each score taken in duplicate. Change in righting reflex reflects the difference between the monitoring timepoints at 18 HPC and 24 HPC.

S4 Figure. Evaluation of baseline and ensemble algorithms in their ability to classify survivors and non-survivors in a neonatal mouse model of polymicrobial sepsis. Algorithms were trained on a set of 148 pups and tested on another set of 74, the accuracy (number of correct classifications over total number of classifications made) is shown on the y-axis.

S5 Figure. 10-Fold cross validation of the Gradient Boosted Machine algorithm.

S1 Table. Accuracy of different algorithms and feature selection approaches in classifying mice into survivors or non-survivors at 24 HPC.

S2 Table. Confusion matrix showing Gradient Boosting Machine algorithm applied to external dataset. Confusion matrix demonstrating the predictive capacity of the classifier when applied to a never seen dataset of 21 neonatal mice challenged with the cecal slurry model of sepsis. The classifier accurately categorized 9/11 survivors and 9/10 non-survivors, in line with the internal cross-validation set.

S3 Table. User-defined hyperparameters to be tuned for each algorithm. Hyperparameters must be set prior to initiating the training process and are critical to constructing a robust classifier.

S1 URL. Public GitHub repository containing the code used to construct the classifiers.

https://github.com/radaniba/Sepsis_Project

S1 Appendix. Classifier construction and background

Our goal was to construct a classifier which could predict whether mice would survive or die 24 hours post challenge (HPC) in a cecal slurry model of neonatal sepsis. The 24-hour time point was selected as it was the final monitoring point before many mice began to succumb to the infection, and thus represented the largest likely gap between the two groups of interest (survivors and non-survivors). As this dataset was assembled over more than a year, the first step was a cleaning process which consisted mainly of removing all data columns that were not relevant to the model training, as they were derived from experiment notebooks and were mainly used for tracking purposes. These attributes were mainly pup identifiers and dates of birth, litter identifiers *etc.* The remaining attributes were only those which could be potentially relevant to the prediction of outcome – this was the starting point for feature selection in our model building. The dataset was a mix of continuous (i.e. weight, hours post challenge) and categorical (i.e. sex, righting reflex) features. Categorical features were encoded as binary operators (1 or 0) using the dummy variable function in the Pandas package – this is required for some of the approaches we pursued.

The selection of the most relevant features to this classification problem is an important step and can be the difference between success and failure. Inclusion of irrelevant features may inject noise into the system and can reduce the overall accuracy. Here we deployed three approaches to feature selection: correlation-based selection, hypothesis testing selection, and machine learning selection. Briefly, correlation-based approaches compare correlation coefficients to identify those which are linearly dependent with one another and are therefore not necessary to include moving forward. Hypothesis testing is a more rigorous approach which statistically tests each feature and returns p-values indicating the likelihood that a given feature is relevant to the outcome of interest, in this case survival. Machine-learning selection relies on a supervised machine-learning algorithm (in this case we used Support Vector Classification) to rank the relevance of each feature to the desired outcome.

Algorithms

As the working dataset of 222 pups was small by data science standards, we were able to test multiple approaches without concern for time or computing power. Baseline algorithms are advantages in their ability to be understood and explained, whereas ensemble learning approaches act more as a ‘black box’ with the line from input to output becoming very difficult to follow. Of course, ensemble learning approaches generally have a higher capacity to model complex problems. Thus, in order to get a sense of which approach was most appropriate for this specific classification problem, we opted to test three baseline learning algorithms and three advanced ensemble learning algorithms.

We selected logistic regression, K Nearest Neighbors (kNN), and simple decision trees for our three baseline learning approaches. Logistic regression has the distinct advantage of being extremely simple to interpret – the output is a probability so it can be used not only in capacity to predict but also to rank confidence of each prediction. However, it is strongly susceptible to overfitting due to multicollinearity (feature correlation with one another) which was clearly present in this dataset (i.e. weight at 0 HPC is strongly related to weight at 18 HPC and weight at 24 HPC). kNN operates under the principal that similar treatments on different subjects will lead to similar outcomes; the principal advantage of kNN is that it requires no assumptions about the data and is non-parametric in its approach. The only information which must be user specified is the proper estimation of ‘k’. It is, however, highly sensitive to outliers and features in the dataset which may be irrelevant. kNN technically does not “learn” in the classic sense, rather it stores data and examines the similarity or proximity of new instances against those already stored. The final baseline approach we explored were simple decision trees, which are generally simple to interpret but even more so when the classification problem is binary in nature. Contrary to logistic regression and kNN, decision trees are insensitive to the linear separability of the data and thus there is less concern regarding outliers and multicollinearity. On the downside, decision trees are highly sensitive to small changes in user input data which can lead to large changes in the trees. Exposure to unexpected data is generally not well handled and can lead to errors in classification.

The more advanced, ensemble learning techniques we used were Random Forest, Gradient Boosting, and XGBoost algorithms. These ensemble techniques merge several weak learners into a single strong one, reducing bias and variance considerably in supervised learning use cases. The specific benefits and downsides of each of these algorithms is beyond the scope of this work. They are generally fast, user-friendly, and well-capable of handling mixed data types (categorial and continuous). These strengths though also make them prone to overfitting and noise, which is why it is extremely important to cross-validate with rigor. Each of these algorithms comes with a set of user-defined hyperparameters which must be set prior to beginning the training, and optimizing these parameters is a critical step in attaining accurate classifications – the parameters are listed in S3 Table. We optimized these parameters using the GridSearch algorithm, which tests all possible combinations of values and returns the most effective values for a given model. For precise parameters used in each case, refer to accompanying code linked in the supplementary materials. The strength of each algorithm was assessed internally using 10-fold cross validation wherein the model is trained and tested iteratively on different subsets of the data. The strongest approach (Gradient-Boosted machine learning) was then tested on an external subset of data which had not been seen by the model.

Feature selection

As feature selection is a widely discussed subject in machine learning, we have different ways of selecting a subset of features that are the most relevant to our classification problem. Using irrelevant features might make the learning process longer and more expensive in terms of resources and could also make the final prediction not accurate and biased because of noise added to the model training.

Correlation based feature selection

In this paper we used Pearson correlation to compare the features. Highly correlated features were more linearly dependent and hence had almost the same effect on the dependent variable (in our case, the outcome being the life or death of the mouse): we then might drop one of the correlated features as there will be no effect on the final outcome.

Hypothesis testing feature selection

We used hypothesis testing as a more complex statistical approach to select the best features from our dataset using a more rigorous feature choice by looking at the p-values. The main purpose of choosing this approach was to examine whether or not certain conditions could be applied to an entire population, from a data sample. The result of a hypothesis test determines if we should believe the hypothesis or reject it for an alternative one. To select the best features in our dataset we were interested in testing the hypothesis of a given feature having no relevance to the response variable. We wanted to test this hypothesis for every feature and decide whether the features hold some significance in the prediction of the response. Based on the p-value per feature, the general rule was that the lower the p-value, the better the chance that we can reject the null hypothesis. In our case, the smaller the p-value, the better the chances that the feature had some relevance to outcome and it should be retained moving forward.

Feature selection using a machine learning algorithm

We investigated the possibility of using a machine learning algorithm as a proxy for selecting a subset of features for the rest of the analysis. Every machine learning approach has an internal logic for ranking features, in this case we used Support Vector Classifier as a method generally recommended to perform for ordinal classification (taking in consideration that outcome was transformed into binary values. Because we are using a machine learning model to rank features, we have an additional complication which consists in the fact that we have an additional hyperparameters to optimize. To tackle this issue, we use the same principle with GridSearch algorithm and combining the feature selection and the training phase optimization within the same pipeline (see details in the notebook attached)

Choosing the best model

In order to validate the use of righting reflex and mobility as metrics of health scores in neonatal mice, we used the machine learning approach described above to classify mice as survivors or non-survivors based primarily on righting reflex, mobility, and weight, specifically looking at how each metric changed from 18 to 24 HPC. Feature selection and classification was performed on components of the scores (righting reflex and mobility) rather than directly on the scores themselves in order to independently quantify the significance of each metric, in comparison to other commonly collected biometric data (i.e. weight). Only 12% (28 / 222) of mice were at humane endpoint at 24 HPC, so while the association between score components and outcome may be influenced by the inclusion of these mice (mice which were FTR non-mobile on both sides will, by definition, not survive) they did not represent a large enough population to call into question the results.

Some of the features describing the dataset were categorical. Because some algorithms are known to be dealing with this kind of feature types better than other (like decision trees for example), we decided to encode these categorical features into numerical ones, using the one-hot encoding technique, thus creating a numerical fingerprint per training data point. For instance, the feature ‘change_righting_high’ describing (….) has categorical values such as ‘ftr.to.rights’ or ‘rights.to.ftr’. When the encoding is applied this feature is then transformed into two newly engineered ones ‘change_righting_high.ftr.to.rights’ and ‘change_righting_high.rights.to.ftr’ with binary values (0, 1). Following this feature engineering process and data cleaning and standardization, we examined the correlation between these attributes and the final outcome and decided to select the features having a correlation with an absolute value higher than 0.2 as a threshold. The heatmap shown in S3 Fig is a graphical representation of Pearson correlation applied to the feature matrix.

Except few attributes highly correlated, we can observe a certain attributes independence, which could be beneficial for the final accuracy of the trained model. Highly collinear features may be a bias in the training phase, as they describe the data point the same way and bring no variability to learn different patterns which may cause either overfitting or underfitting the model to the training set. In order to have the optimal model, we need to account for two parts. First, we need to make sure that we are using the attributes that describe the best our data patterns, and second, we need to use the best algorithm with these selected attributes. Thus, we have a combination of six algorithms and three feature selection methods. To add some complexity on the mix, every algorithm comes with different parameters that we need to play with to tune the model, luckily, we have the Grid Search method which is a hyperparameter optimization process that will test all possible combination of values and give us the best values to use for the model. We performed a 10-fold cross validation training using all these items described above and we measured the training accuracy for each use case. The result is displayed in S1 Table.

We can clearly see that, for our small dataset, the accuracy is high for all the algorithms used (> 0.8) which is a very good indicator that the problem we tried to solve (predicting mortality) is predictable given the data set in hand. We need to be sure about the optimal solution though by minimizing the error rate as much as possible in order to inform accurately on the subject we need to use the biological material from subsequent experiments. Gradient Boosting and XGBoost are two algorithms that, combined with feature selection based on correlation, gave the best training accuracy compared to the other methods (S1 Table). We thus decided to select Pearson correlation and these two ensemble learning to build our optimal solution. S4 Fig shows the training accuracy per Classifier family. We decided to use the Gradient Boosting Machine as our classifier combined with features selected by Pearson correlation. The S5 Fig below describes how this accuracy is reached.

Testing the Gradient Boosting Machine model

With the hyperparameters selected for our Gradient Boosting model, we built a model on the training dataset and applied it to a 33% of the original data set not used during the training phase. The goal is to test whether or not our model is generalizable. With a total of 74 cases to classify as “Live” or “Die”, our model using Gradient Boosting performed extremely well with an accuracy score of **0.85** and an ROC value of **0.93**, with 33 True Positive cases classified as “Die” and 30 cases classified as “Live”, the confusion matrix shown in S2 Table shows the accuracy of our model. To further validate our model, our team generated an additional dataset with the same data collection process as well as the same data cleaning and standardization pipeline. With a total of 21 new data points, our model was able to accurately classify 18 with an average score of **85% accuracy** (S2 Table).

S2 Appendix. Reference guide for associated data

Data file overview

mouse_mortality_data.xlsx [266 mice]

This is the data set used for all the explorations into score, righting reflex and mortality. All the mice from classifier_building_data.xlsx and classifier_validation_cohort.xlsx (combined 243 mice) are represented here. The additional 23 mice were excluded from model construction due to incomplete data collection (i.e. missing weight, monitored at 21 hours and 30 hours rather than 24 hours, etc.) but were still able to be used in things such as survival curves.

classifier_building_data.xlsx [222 mice: subset of mouse_mortality_data.xlsx]

This was the data used to train and internally test the model. It is presented separately as these were the features that were presented as the starting point for classifier construction. For internal cross-validation, this set was further split into a training set (66%, 148 mice) and test set (33%, 74 mice).

classifier_validation_cohort.xlsx [21 mice: subset of mouse_mortality_data.xlsx]

This was the 21-mouse external validation cohort used to generate Figure S3 and validate the classifier. As above, it is a subset of mouse_mortality_data.xlsx filtered to only include features which were presented to the classifier.

mouse_CFU_data.xlsx [125 mice]

These mice were all sacrificed at 24 HPC, dissected to calculate bacterial load in blood, spleen, liver, lung, and then classified into predicted survivors or predicted non-survivors.

sham_challenge_control_set.xlsx [33 mice]

These mice were included exclusively to show the morbidity and change in score was not associated with the needle poke. Used only to generate supplementary figure 1 (S1 Fig).

~~Supporting Information~~

~~**Figure S1. Distribution of scores assigned to neonatal mice at 18 and 24 hours post challenge.**~~ ~~Scores assigned to neonatal mice 18 and 24 hours post IP challenge with cecal slurry (HPC). At 18 HPC the scores are poorly distributed, with the vast majority of mice assigned a score of 2 (failure to right, mobile) indicating that most mice have not progressed towards survival or non-survival. By 24 HPC the scores are evenly distributed as mice have begun to succumb to or recover from sepsis.~~

~~**Table S1.** Accuracy of different algorithms and feature selection approaches in classifying mice into survivors or non-survivors at 24 HPC.~~

~~Method~~	~~**GridSearch (No feature selection)**~~	~~**Correlation + GridSeach**~~	~~**Hypothesis + GridSearch**~~	~~**Model based + GridSearch**~~
~~**Logistic Regression**~~	~~0.860~~	~~0.874~~	~~0.874~~	~~0.874~~
~~**K Nearest Neighbor**~~	~~0.827~~	~~0.833~~	~~0.851~~	~~0.856~~
~~**Decision Tree**~~	~~0.820~~	~~0.842~~	~~0.842~~	~~0.869~~
~~**Random Forest**~~	~~0.878~~	~~0.869~~	~~0.874~~	~~0.878~~
~~**Gradient Boosting**~~	~~0.883~~	~~**0.887**~~	~~0.887~~	~~0.883~~
~~**XGBoost**~~	~~0.883~~	~~**0.892**~~	~~0.892~~	~~0.887~~
~~**Average**~~	~~**0.859**~~	~~**0.866**~~	~~**0.870**~~	~~**0.875**~~

~~**Figure S2. 10-fold cross validation of Gradient Boosting Machine algorithm.**~~

~~**Figure S3. External validation of Gradient Boosting Machine algorithm.**~~ ~~(A) Confusion matrix demonstrating the predictive capacity of the classifier when applied to a never seen dataset of 21 neonatal mice challenged with the cecal slurry model of sepsis. The classifier accurately categorized 9/11 survivors and 9/10 non-survivors, in line with the internal cross-validation~~

~~set. (B) Confidence of predicted outcome, survivor or non-survivor, within the 21-mouse external validation set.~~

~~**Figure S4. Feature selection visualization using Pearson correlation.**~~

~~Heatmap of feature correlations of with Pearson correlation. Scores were split into components, starting with looking at righting reflex and mobility independent from one another and then further separated by the lower and higher measurements of each score taken in duplicate. Change in righting reflex reflects the difference between the monitoring timepoints at 18 HPC and 24 HPC.~~

~~**Figure S5. Distribution of training accuracy per classifier family**~~

~~Classifier construction~~

~~We collected the experimental data per mouse and mapped it to the outcome of the results at 24 HPC. Outcome categories were defined as survivors and non-survivors. The raw data collected was cleaned and preprocessed before building the model. Attributes that were not useful for classification, such as unique identifiers, dates and times, were removed. The final dataset was well-balanced, containing 112 non-survivors and 110 survivors. Following feature cleaning and engineering (i.e. separating scores into categorical components) the dataset had 23 attributes. We tested and compared six different algorithms which fit within two distinct approaches; baseline learning algorithms (Logisitic Regression, K Nearest Neighbors and Decision Tree) and ensemble learning algorithms (Random Forest, Gradient Boosting and XGBoost). Feature independence was evaluated and visualized using Pearson correlation coefficients (Fig S4).~~

~~Except few attributes highly correlated, we can observe a certain attributes independence, which could be beneficial for the final accuracy of the trained model. Highly correlated features may be a bias in the training phase, as they describe the data point the same way and bring no variability to learn different patterns which may cause either overfitting or underfitting the model to the training set. We tested three methods of feature selection (Pearson correlation, *K* nearest neighbor, support vector classification) and Grid Search optimization. We performed 10-fold cross validation training using all aforementioned items measured the training accuracy for each use case (Table S1).~~
~~The accuracy was is high for all the algorithms used (>0.8) which indicates the prediction of outcome to be a solvable problem. Gradient Boosting and XGBoost with Pearson correlation as feature selection~~

~~gave the best training accuracy compared to the other methods. We thus decided to select Correlation and Gradient Boosting (Fig S5).~~

~~There was a convergence to a cross validation average accuracy of 0.91 with a low standard deviation of 0.06, indicating high accuracy with little risk of over or underfitting. With the hyperparameters selected for our Gradient Boosting model, we built a model on the training dataset and applied it to a 33% of the original data set not used during the training phase. With a total of 74 cases to classify as "survivor" or "non-survivor", Gradient Boosting model performed well with an accuracy score of 0.85 and an AUROC value of 0.93, with 33 True Positive cases classified as "non-survivor" and 30 cases classified as "survivor". The model was further validated with a truly external set of 21 mice wherein the previous accuracy was again seen (0.86) (Fig S3).~~

RESPONSE TO ACADEMIC EDITOR

We appreciate your time and effort in editing this manuscript – we have addressed each point as follows: (1) The manuscript has been edited to ensure it follows all style requirements and incorrectly named files have been renamed in order to meet the journal criteria. (2) The animal care section has been greatly expanded to ensure adequate detail has been provided. (3) Tables in the supplementary material have been uploaded separately as their own files.

RESPONSE TO REVIEWER #1

Thank you for your detailed feedback and critique of this manuscript. We respond to each point raised below.

- *The manuscript lacks sufficient clarity to highlight the work done or the significance of it. The methods need significant work to describe what exactly was done.*

We have expanded the methods section to include a more detailed account of the development of the scoring system in order to highlight how this came about and have extensively reworked the section describing classifier construction. In brief, we developed a scoring system and showed that recording these data can enable investigators to predict outcome of neonatal mice. This opens doors for the investigation of neonatal infectious disease, which is limited by an inability to collect biological samples from neonatal mice without sacrifice. We accordingly have expanded the discussion to include greater emphasis highlighting the significance of the work.

> • *Line 80: remove the duplicated word, "markers"*
> • *Line 87: I think it is incorrect or overstating to state that you validated your health scoring system…*
> • *Line 102-103: What was the volume of the IP injection into the pups?*
> • *Line 103-104: You indicate that the litters were monitored for "morbidity (i.e. humane endpoint)" but do not describe what signs of sickness you are looking for*

Typos in the manuscript have been corrected. The word 'validated' has been changed to 'correlated' to correctly reflect the data. Volume injected (~100 µL, some adjustment for weight) has been added to the methods section. The specific criteria for morbidity / humane endpoint are one of the focal points for the paper – we principally looked at righting reflex and mobility or lethargy. As this is described in much greater detail in the main text.

- *What is meant by the statement "all efforts were taken to minimize animal suffering"?*

As there is no standard humane endpoint or measure of sickness for neonatal mice, we feel this manuscript itself represents the claim that 'all efforts were taken to minimize animal suffering'. We strived to identify the earliest possible point where mice that were destined to not be able to recover from disease could be identified to reduce animal suffering, and for biological importance to identify what makes those mice different from mice that go on to survive but still look sick at the same point.

- *Line 116-117: please be more explicit in your description of the animals included in the analysis – were they all receiving cecal slurry? Did any receive treatment? All part of an LD50 study? All wildtype B6 mice? Did they all go to the humane endpoint or were some euthanized at 24 hours?*

- *Line 120: You indicate that animals were monitored 2-3 times daily throughout the experimental endpoint. What was the specific interval? Did the interval depend on the time after CS injection? What criteria determined when the animals were monitored? What time were the observations at? Light cycle or dark cycle or both?*

We have edited the main text to address these questions and clarify these experiments. In survival studies mice were let to go to humane endpoint or recovery, whereas in experiments with sample collection mice were sacrificed at 24 hours. Specific intervals have been added to the methods as has the time of injection and the clarification that this occurred in both light and dark cycles.

- *Please include a description of the husbandry of these animals. What sort of housing was used? What cage density? What enrichment was offered? What diet and water source were the dams offered? What was the light cycle? What was the temperature and humidity?*
- *Please describe how pups were identified and scored individually? How was this information tracked? Did you use scoring sheets for each animal?*
- *Please describe weight collection method since this data is later used as a metric for endpoint criteria evaluation*

Details and clarifications addressing these questions have been added to the methods section of the text.

- *Please fully describe the statistics that were used to evaluate the data? What test? Was data tested for normality? What program was used to evaluate the data?*

An additional section titled 'statistics' has been added to the methods section in order to address these points. Some of the statistical tests used throughout the paper have been modified to better suit the data – the results have been altered accordingly.

- *Figure 1: I don't understand what is plotted in A. What does the total bar at each time point represent? Why does it increase and decrease - I would expect that it would steadily decrease over time.*

The total bar at each point represents the total number of mice monitored within that given bin. It does generally decrease over time, fluctuations are due to differences in monitoring time points (i.e. not many mice are monitored at 27 hours because they were just monitored at 24 hours)

- *Figure 1C: were these animals all challenged with the same dose of bacteria?*

All animals were challenged at a cecal slurry concentration corresponding to an LD50 – this varied slightly between different preparations of cecal slurry (from 0.8 mg-1.1 mg). The caption has been updated to better reflect this information.

- *Table 1: How did you determine a 4 second cutoff time for the righting reflex? This seems like a very long time to wait for the animal to do this?*

The 4 second cutoff was determined by observing the natural righting reflex of non-challenged pups, and picked as a time when over 95% of mice were able to right on at least one side after being placed on their back. Any shorter and healthy mice would display fail to right reflexes, while longer could lead to exhaustion of the mice which could potentially affect the response to disease.

- *Table 1: While it is easy to understand a present or absent righting reflex, I do not follow the logic in the organization of the Mobile/Lethargic/Nonmobile associated with the scoring? Can you please explain this in the methods section – how you devised the complete scoring system.*

It is difficult to give a detailed account of the development of the scoring system, as this arose from a large number of qualitative observations over the course of multiple years of running neonatal mouse experiments. The need for consistency and the ability to predict outcome drove us to turn these qualitative observations into a quantitative measurement, hence this scoring system has emerged. We have attempted to provide some background and context to address this in the methods section, and have expanded the previous discussion to include greater information about this mobility metric.

- *Figure 2A: What does the shaded area represent?*

The shaded area represents standard error – this has been added to the figure caption.

- *Figure 2B: What is plotted in this graph and what does the shaded area represent? Are these averages or modes at each time point? A fitted model? You stated that the scores were discrete categorical data but you have a smoothed continuous curve – please explain. Can you show the actual time points for which you collected this data?*

The lines represent loess fits applied to the scores treated numerically and the shaded area represents standard error. We did it was inappropriate to treat the scores as numerical variables for the purposes of statistical analysis as it is impossible to claim that the difference between each category is precisely the same arbitrary health unit. However, we felt that treating the scores as numerical entities for the sole purpose of visualizing the change in score over time was acceptable as there is no statistical test or weight being assigned to the results. We attempted to emphasize this by specifying it was a 'visual representation' – the basis of our claim that 'score changes over time' can be seen in figure S1 where the scores were not treated numerically.

- *Lines 187-193: It may be useful to use a ROC curve and Youden's index to evaluate the combination of parameters you use as a cutoff – this will optimize the sensitivity and specificity of the combination of score and time point for predicting death and thus an ideal endpoint for this model.*

We appreciate this suggestion for evaluation of the timepoint for predicting death, and we agree that there could be some improvement selecting the best time to predict death. However, we are somewhat limited in this manuscript by the data we have available – as most of this data was accumulated over time naturally as a result of other experiments in our lab, we were not increasing our monitoring time points to seek out the data to optimize this. So, we ultimately were presented with the choice of 12, 18, or 24 hours and here we demonstrated that, of these choices, 24 hours was the best candidate. This has been added to the discussion as a limitation of the study.

- *Figure S1: Is this for all mice at 18 and 24 hours?*

This is for all mice with known outcome (i.e. were not sacrificed) at 18 and 24 hours. Clarified in figure caption.

- *Line 200-201: How many mice were used? Where these mice also scored using the criteria in Table 1?*

125 mice were used, this information was added to the text. These mice were scored using the same criteria.

- *Line 204: please clarify if the bacterial load is only assessed at 24 HPC.*

Clarified in text

- *Line 209-210: What data supports this statement? What health score was associated with survival? How was it correlated to bacterial load?*

This text is supported by all of the data presented in previous section – Figure 2A shows a clear relationship between score at 24 HPC and percent survival and Figure 3 shows a direct correlation between increasing score and decreasing bacterial load (measured at 24 HPC) at both 18 and 24 HPC. We feel this data is clearly presented in the section.

- *Line 222-223: You indicate that you use righting reflex and mobility to make a single score but then use them separately here and weight with the model. How was this information handled? Why did you separate them rather than keep them together?*

Our logic in separating the score into individual components was that we wanted as close to a truly unsupervised classification process as possible – by using our created scores as inputs we worried we would limit the discriminatory capacity of the classifier. The scores were deconstructed into their smallest possible components. Mice were placed on both sides and righting reflex and mobility was collected twice so there were a total of four measurements comprising each score. Our goal was to discriminate survivors and non-survivors as effectively as possible so it logically follows to include as much information as is available. The utility of the scores external to the classifier lies more in the simplicity of collection and interpretation and thus does not make sense to separate into components.

Discussion

- *Can you please discuss why you came up with score criteria but then analyzed righting reflex and body weight separately.*

A brief summary of the previous response has been added to the discussion section.

- *What would you specifically recommend to use as a humane endpoint criteria with this data?*

As we mention in the penultimate paragraph of the discussion, we suggest that mice which fail to right and are lethargic should qualify as humane endpoint in future studies.

RESPONSE TO REVIEWER #2

We appreciate your response and critique, especially of our classification section. We have heavily modified the section to address your concerns – each point is addressed below.

> *The limitations in our knowledge about the way in which newborns' immune systems handle infectious threats such as those that might lead to sepsis is correctly identified and the use of mouse models seems, in principle, like an adequate route towards increasing such knowledge. Authors define a standardized metric to quantify neonatal mouse health in its relationship to outcome (adult mouse metrics are available but not readily applicable to newborn animals. The reported interest of this metric is that it is shown to be highly predictive of survival (understood as a classification problem).*
>
> *The paper is well written and reasonably well structured, and the reported research falls adequately within the scope of the PLoS ONE journal.*
>
> *The concept of sepsis is introduced in the first section without any basic explanation or definition. Given the generalist nature of the journal, such explanation/definition would be welcome.*

A sentence overviewing the concept of sepsis has been added to the introduction.

> *The description of the data analysis methods (section "Classifiers" and supplementary material) is very insufficient. Authors state that they "built a machine learning model for a binary classification problem", but the fact is that only part of their classifiers could be considered to be machine learning models. More importantly, there seems to be no clear justification or rationale behind the specific choice of algorithms. Such justification should be made explicit in the text.*

We have heavily modified the classifier section and re-wrote the supplementary material entirely to try to improve clarity and provide more background that may be required for this manuscript to flow naturally. The new supplementary section is primarily focused on the differences between the approaches and the logic behind our choices. Additionally, we have created a public GitHub repo wherein the code can be reviewed which should clarify any specific questions (S1 URL).

> *All these algorithms depend on a number of parameters that must be tuned to create the classifier model. No reference is made to how the optimal values for these parameters were selected. Without such information, the experimental results have no guarantee of replicability. Authors should also clarify how they tackled the potential problem of overfitting in their models. The evaluation metrics (accuracy and AUC), even if standard, should be formally described and justified. The feature selection methods are barely described (for instance, what is KNN and SV classification feature selection?)*

We have added an additional table in the supplementary information which outlines the user-defined parameters which needed to be defined in order for the algorithms to progress. We principally used the GridSearch technique, which we have now defined and included in the manuscript. Further, we have added a discussion of the cost / benefits of each algorithm and provided some rationale behind the choices made. Additionally, we have uploaded our Python workbook which details the whole code and should address any concerns about repeatability. We have also now clarified accuracy and AUC in the main text.

In the discussion, I miss a couple of things: 1) comment on the relevance of feature selection as part of the classification analysis and as a way to make the interpretation of the results more accessible to medical experts; and 2) some comments on how this preliminary research on mouse models could eventually lead towards an approach that might be useful in human neonatal research.

The relevance of feature selection has been added to both the results and the discussion, emphasizing the significance of righting reflex and its quality as an easy to use, predictive metric. We have also added some additional lines in the discussion highlighting the relevance to human neonatal infectious disease.

Figures:
All figures seem to be of rather low resolution (don't know about the originals, but they look so in the pdf file)

Resolution has been fixed on figures and some have been removed.

Confusion matrices in Figs.4 and S3 do not make much sense as figures. They would be better reported as tables.

The confusion matrices have been replaced with tables.

Several figures in the supplementary material come with no descriptive text.
I do not quite understand what the horizontal axis represents in Fig.S2

We have updated some figures which erroneously did not include axes labels. Per the PLoS ONE style requirements, supplementary figures are not required to include descriptive text – those which do not have this text are referenced in the supplementary materials and are meant to be understood in the context of that document.

Data are adequately made publicly available from the Open Science Framework database.

RESPONSE TO REVIEWER #3

Thank you for your detailed review of our manuscript – your feedback has been very helpful in guiding us to clarify some confusing points. We address each of your concerns below and have added additional supplementary information.

Neonatal mouse could be utilized as a model and provide valuable data for the research of sepsis in newborn babies. Due to the significant hardness in collecting biological samples in neonatal mouse without sacrificing the animals, new methods based on easy-obtained quantitative and qualitative metrics beyond the regular adult predictors are essential and should be established.

In this manuscript, the authors successfully presented some critical new metrics and constructed a scoring system to indicate the health and bacterial load of neonatal mouse. In addition they constructed a classifier to predict the mortality and validated the robustness.

Generally this research is quite innovative and produced good and meaningful results.

But there are some questions/problems to be answer/clarify and some flaws to be corrected:

1) Authors should demonstrate the bacterial load but not the IP injection surgical damages and related stress is the main reason of changes in righting reflex.

We have inserted an additional supplementary figure showing that sham challenges (injection of either dPBS or dextrose 5% water) did not cause any mortality (A) and had significantly higher scores than mice which received a cecal slurry challenge (B).

2) Prediction accuracy alone is NOT a good indicator of prediction performance as it's hardly to be extended to the usually highly unbalanced real world data.. Authors could replace it with (macro or micro) F1 score, because it much more robust.

We entirely agree with the reviewer's sentiment that accuracy is a poor metric of performance in real world (unbalanced) data, but we believe it is appropriate in this context where we do in fact have a highly balanced set (112 survivors and 110 non-survivors). Still we did look at F1 scores and found virtually no difference compared to accuracy – these can be seen in the now posted code associated with classifier construction. Again, we entirely agree with the reviewer, but our inclination is to keep accuracy as the presented metric in the paper for simplicity and readability, as the intended audience is experimental scientists rather than experts in machine learning. This classifier is not intended to extend to real-world data - its utility lies in the ability to discriminate survival in a controlled laboratory setting where it is unlikely to encounter highly imbalanced groups.

3) Line 173:
Authors should show all data about the timing of mice death, no matter what's the prediction results. If the death was long after the scoring, it's could caused by factors other than the infections and it hard to validate effectiveness of scoring/prediction.
The autopsy data could validate the method if they are available.

We agree with the reviewer that all deaths are important regardless of the timing, and for all statistical analyses (and classification) we indeed included all deaths. Specifically for Figure 2B, the loess fit shown no longer makes sense as the data gets extremely sparse after 48 hours. The monitoring time points become less frequent for the survivors (as they are already in recovery) and so the majority of the recorded scores are mice which continue on to death, as these mice are still being monitored frequently. All mice have reached recovery or death by 96 HPC so it seems unlikely to us that these few deaths after 48 HPC are unrelated to the infection. Unfortunately the autopsy data are no longer available for this dataset.

4) Line 230:
Authors should clearly demonstrate what's the initial features and the rational of the initial features composition.

An overview of the feature inclusion has been added to the methods section – the rationale was to include all information available that was present in this data set and then employ a rigorous, data-driven method of selecting the most relevant features. This data set is comprised of mice from many experiments over a multi-year span which happened to be treated the same (i.e. received a cecal slurry challenge at the correct dose and were not exposed to any sort of treatment or intervention). This paper and classifier represents us working with the data which we had been collecting already rather than an experiment-driven approach to defining neonatal mouse health. We realized these data we *had* been collecting were very strong in of themselves and so set out to create a scoring system which can be adopted by other labs working with neonatal mice. So in summary, the rationale behind the initial feature selection is that these were things we had qualitatively observed to be relevant to survival and therefore were worth collecting. We have attempted to make this more clear in the manuscript.

> *5) Line 233:*
> *Why Pearson correlation coefficients were used to select features and what's the threshold ? why choose the threshold?*

Information about the threshold (0.2) has been added to the methods. Similarly to hyperparameters optimization, when we do correlation and decide to select feature we need to have a method that "tells us" or suggest us the threshold to use to have, along with the hyper parameters optimized, the best combination for the best model accuracy, thus the choice of the best cutoff to use to pick our features is optimized and not based on an arbitrary cutoff choice.

> *6) Line 237:*
> *Why only 74 cases to be used to classify as survivor/non-survivor? In fact the number changed everywhere (in line 117, in "CFU_with_predicted_outcome.csv", in "Mortality_data_training_and_test.csv", and in Fig 1B -- total mice is 138+128 = 266)*

We have attempted to clarify the confusion regarding the mouse numbers by overviewing the numbers more explicitly in the methods section. Out of 266 mice, 21 were saved for an external validation and 23 had missing values that made them ineligible to predict outcome. These mice were still able to be used in the direct comparisons of score / righting reflex and mortality. 222 mice were used to train and test the classifier, which meant these were further subset into to categories: 66% were for training (148) and 33% (74) were for testing. This is where the 74 comes from. We have also added another supplementary document outlining the datasets and the number of mice within each in order to help alleviation any potential confusion.

> *7) Line 247:*
> *What's biometric data? Authors should clarify.*

Biometric data meaning weight and sex but principally weight – we agree this is a confusing term and have adjusted the figure title accordingly.

> *8) Line 257 and line 275:*
> *The surgical damage and operation stress could cause morbidity, authors should exclude the effects of surgical damage and operation stress or at least clearly discuss it in detail.*

We have addressed this with an additional supplementary figure (discussed in answer to question 1).

> *9) Line 392:*

Where is the details/setting/parameters about the Method: GridSearch and combination of GridSearch with others methods?

We have now uploaded the code used to construct this classifier and have directed readers there if they would like to see the specific parameterization used. We agree that these parameters are critical for repeatability, but we feel it is beyond the scope of this paper to discuss the settings of GridSearch in detail given the intended audience of experimental scientists.

10) In the data of "CFU_with_predicted_outcome.csv"
10.1) Why there are only 164 mice info? It's different from 404 mice authors claimed in line 117
10.2) Why are there many NA/missing value in the column "Probability.Live"?
10.3) What's the method to produce the data in the column "Probability.Live"?
10.4) Irrelevant data (columns) should not be included

Thank you for your feedback on these data. This dataset represents a subset of mice which were sacrificed at 24 hours and the bacterial load collected. The missing values in probability live were mice that were erroneously included in this data set and were not represented in the paper – these mice have been removed and irrelevant columns have been cleaned. To alleviate potential confusion, we have added an additional supplementary document clarifying the contents of each file which has been uploaded. We have also added a glossary tab to each data frame attached to the manuscript, per your later suggestion.

11) In the Fig. S4:
11.1) Why "v18.hour.post.challenge" "v24.hour.post.challenge" used to check correlation? It looks pointless
11.2) What's the meaning of "chal.status_non_challenged" ?
In fact, every headline column/item should be explained or have glossary of the definition,
11.3) The rationale these items are included should be explained

These columns show the exact time post-challenge that mice were monitored. All features were checked for correlation to ensure an unbiased assessment – we agree it was highly unlikely that the very slight differences in time post challenge would make a difference, but it is important to not make assumptions when it can be easily confirmed by the machine. This indicated whether mouse received a cecal slurry or sham challenge – two mice in the set were unchallenged and not excluded. A glossary has been added to an additional tab in each data set to clarify the meaning of column names and values within.

12) In the data "Mortality_data_training_and_test.csv"
12.1) Why there are 299 mouse info? It's different from 404 mice author claimed in line 117
12.2) how to treat or imputed the missing data in the above tables?
12.3) Irrelevant data (columns) should not be included.

This data set represents all of the mice for which mortality data was collected – again, some mice were incidentally included which had been filtered out of the final paper due to missing data (hence the prevalence of missing data here). These have been removed and two additional subsets have been added for clarity. Irrelevant columns were again removed.

13) Table S3:

Where is the label of Y-axis in Fig S3 B?
14) Table S5:
Where is the label of Y-axis?

The labels have been added to the axes, thank you.

15) Where is the information about the programming language/ softwares/ libraries ?
16) Where is the programming code?

In response to both questions 15 and 16 - the programming code has been uploaded to a public github repository and a link has been provided in the supplementary information. For ease of access it is also linked here:

https://github.com/radaniba/Sepsis_Project

1) Line 80, two repeated "markers"
2) Line 139, 140, 145 -- Fig 1A, Fig 1b, Fig 1c;
Either up case or lower case, but keep consistent.

Thank you for identifying these errors, they have been corrected in the manuscript.

Data Review URL

Click here to download Data Review URL https://osf.io/r2p9a/files/

S1 Source code URL

Click here to download Data Review URL
https://github.com/radaniba/Sepsis_Project